软 件 工 程 课 程

Chapter 1: 概述

Xi'an Jiaotong University

王 鸽

西安交通大学 计算机学院

gewang@xjtu.edu.cn

Chapter 1 Outline

- Software Development and Systems Analysis and Design （软件开发和系统分析与设计）
- Systems Development Lifecycle （系统开发生命周期）
- Iterative Development （迭代开发）
- Introduction to Ridgeline Mountain Outfitters
- Developing RMO’s Tradeshow Systems
- Where You are Headed—The Rest of the Book

Learning Objectives

- After reading this chapter, you should be able to:
  - Describe the purpose （目的） of systems analysis and design in the development of information systems
  - Describe the characteristics （特点） of iterative systems development
  - Explain the six core processes （6 个核心过程） of the Systems Development Life Cycle

Learning Objectives

- Identify key documents （关键文档） that are used in planning a project
- Identify key diagrams （关键图表） used in systems analysis and systems design
- Explain the utility of identifying use cases （用例） in systems development
- Explain the utility of identifying object classes （对象类） in systems development

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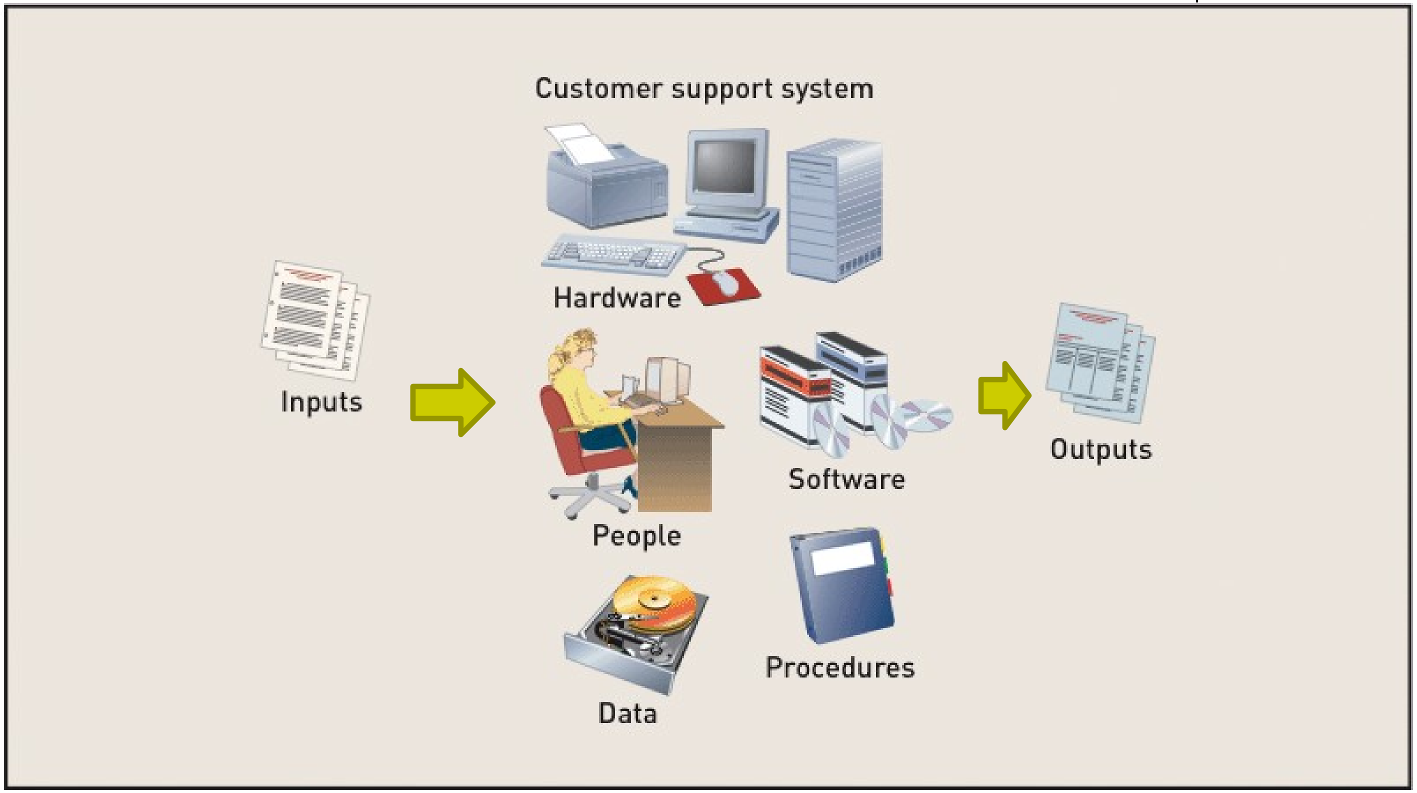
Software Development and Systems Analysis and Design

- **Computer application** (app， 计算机应用程序) – a computer software program （软件程序） that executes on a computing device to carry out a specific set of functions
  - Modest scope
- **Information system** （信息系统） － a set of interrelated components （组件） that collects, processes, stores, and provides as output the information needed to complete business tasks
  - Broader in scope than “app” （范围比 app 更广）
  - Includes database （数据库） and related manual processes （手动过程）

Information Systems and Component Parts

Figure 1-3

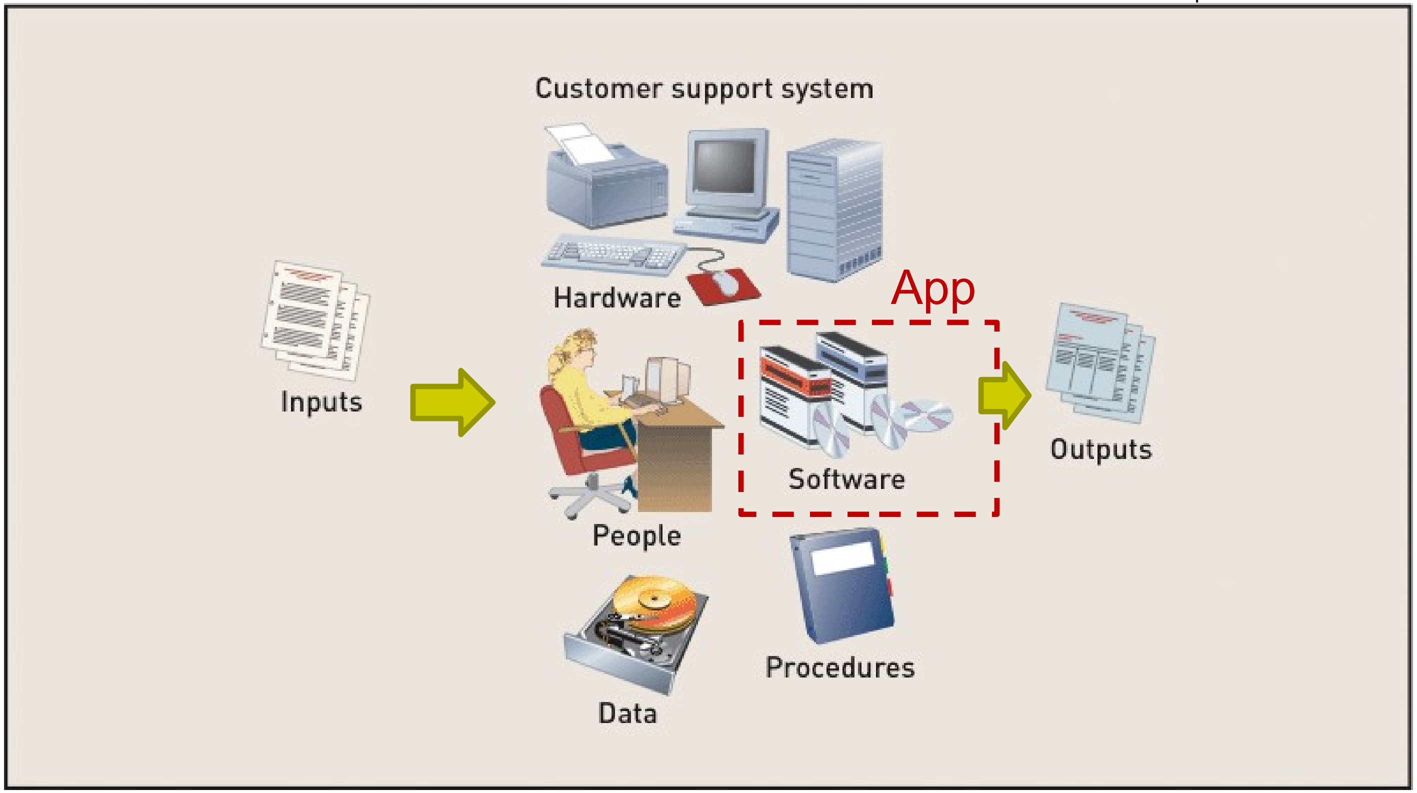
Information systems and component parts



Information Systems and Component Parts

Figure 1-3

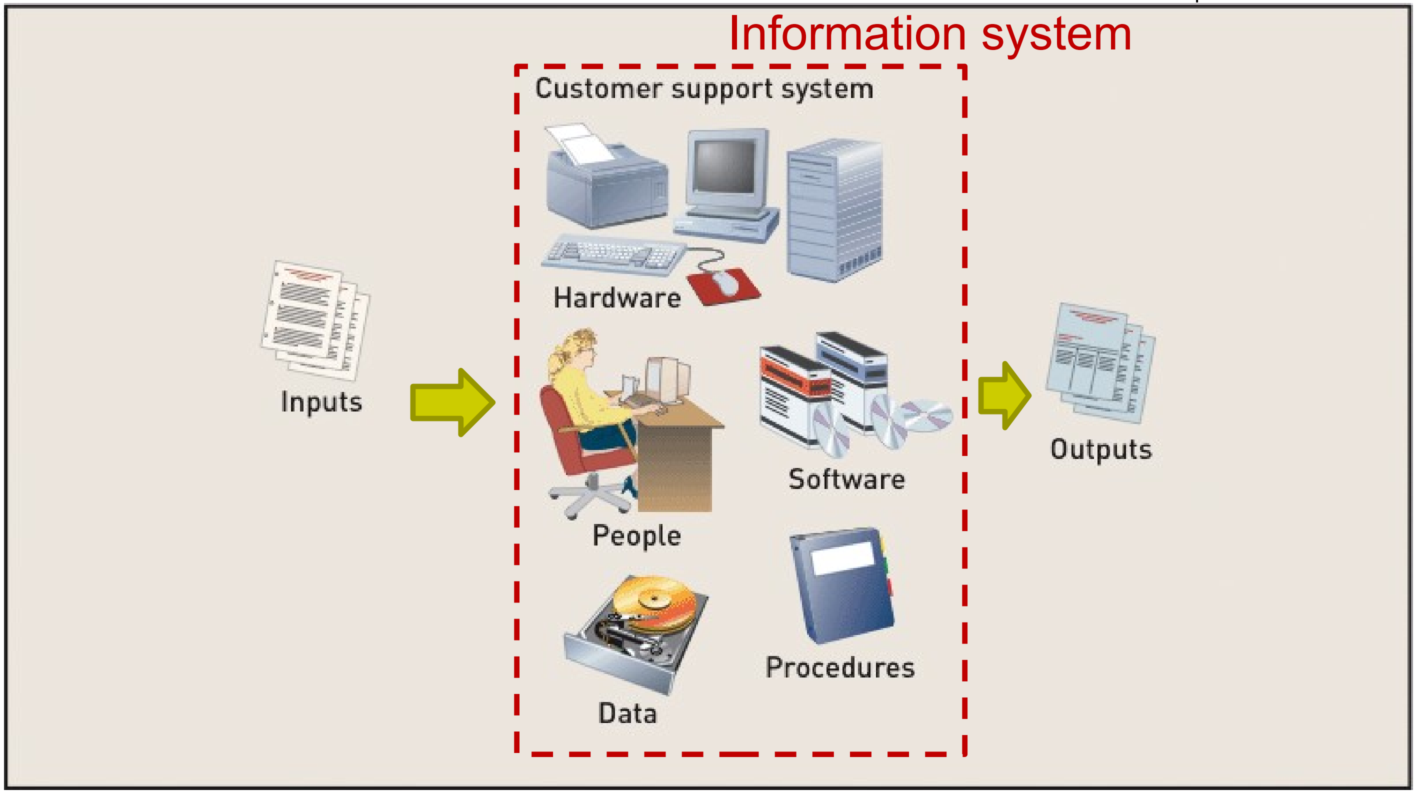
Information systems and component parts



Information Systems and Component Parts

Figure 1-3

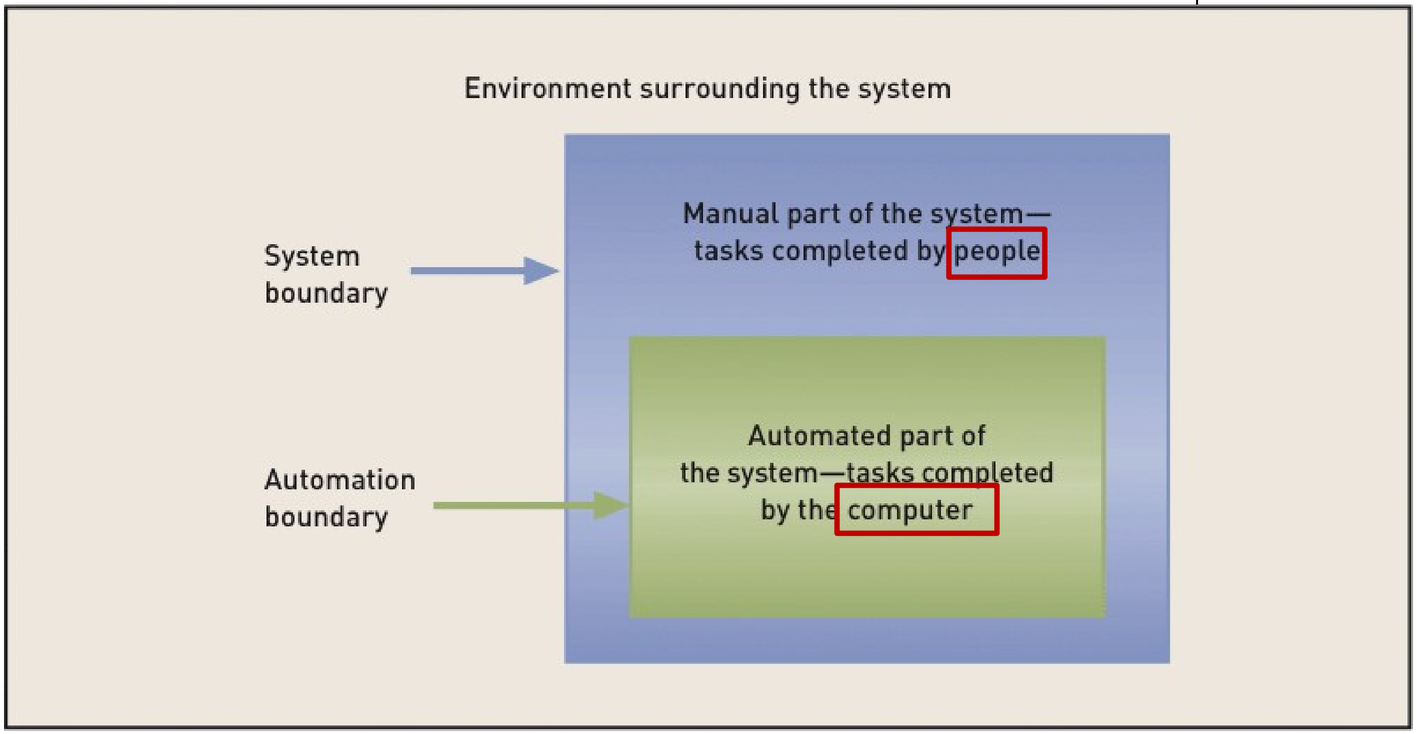
Information systems and component parts



System Boundary vs. Automation Boundary

Figure 1-4

The system boundary versus the automation boundary



Software Development and Systems Analysis and Design

- **Project** – a planned undertaking that has a beginning and end and that produces some definite result
  - Used to develop an information system （用于开发信息系统）
  - Requires knowledge of systems analysis （系统分析） and systems design （系统设计） tools and techniques

Software Development and Systems Analysis and Design

- **Systems analysis** – those activities that enable a person to understand （理解） and specify （指定） what an information system should accomplish
  应该做什么?
- **Systems design** – those activities that enable a person to define （定义） and describe （描述） in detail the system that solves the need
  应该如何做?

The Analyst as a Business Problem Solver

- Has computer technology knowledge and programming expertise （编程知识）
- Understands business problems （理解业务问题）
- Uses logical methods for solving problems （用逻辑方法解决问题）
- Has fundamental curiosity （好奇心）
- Wants to make things better （企图心）
- Is more of a business problem solver than a technical programmer （是业务问题解决者，而不是程序员）

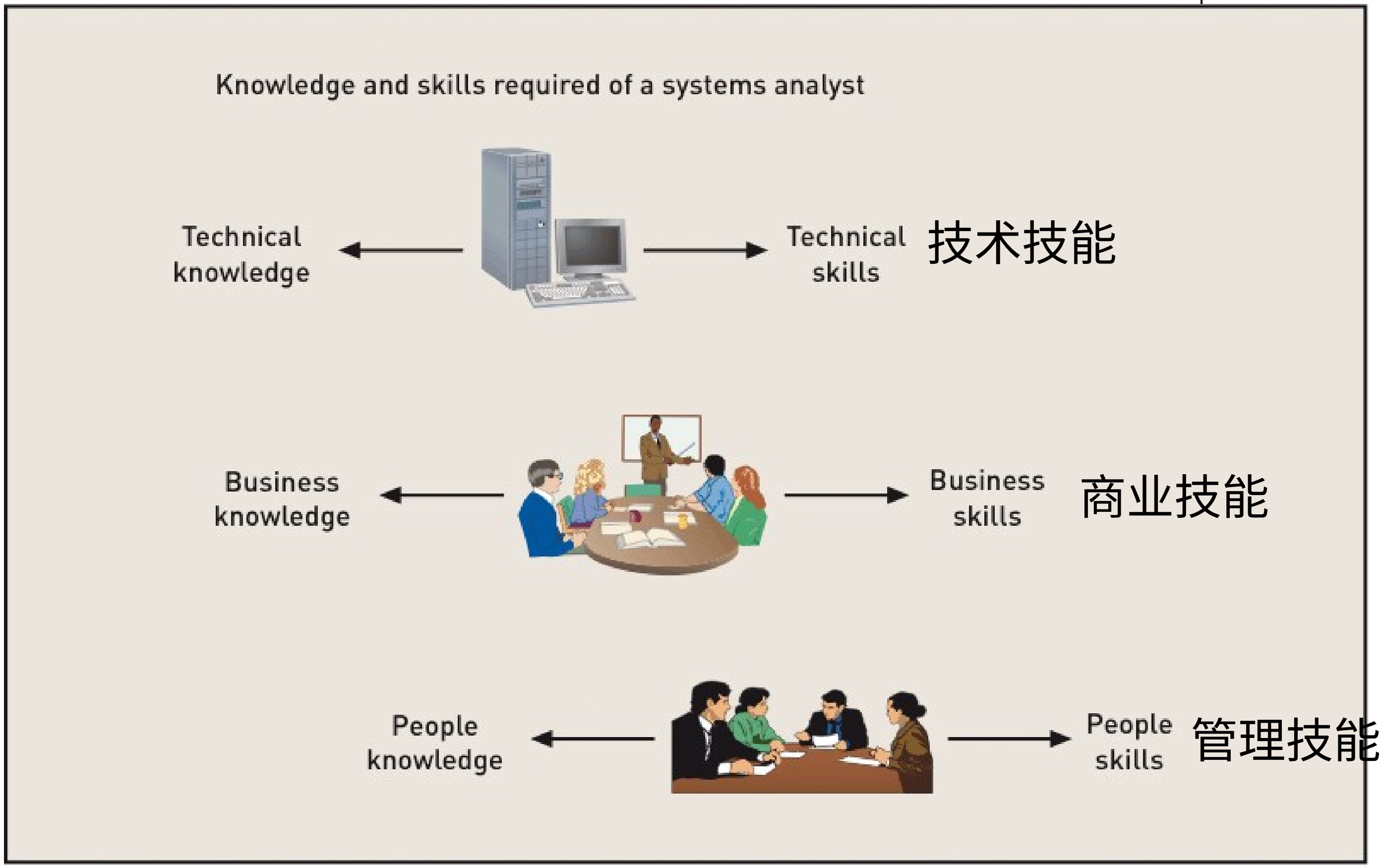
Analyst's Approach to Problem Solving

1. Research and understand the problem （理解问题）
2. Verify benefits of solving problem outweigh the costs (验证收益大于成本)
3. Define the requirements for solving the problem (定义解决问题的所需)
4. Develop a set of possible solutions (提出可行方案)
5. Decide which solution is best and recommend (决定哪个方案最好)
6. Define the details of the chosen solution （决定细节）
7. Implement the solution （实施方案）
8. Monitor to ensure desired results （过程监督）

Required Skills of the Systems Analyst

Figure 1-6

Required skills of the systems analyst



Integrity and Ethics
（诚信与道德要求）

- Analyst has access to confidential information （机密信息）, such as salary, an organization’s planned projects, security systems, and so on.
  - Must keep information private
  - Any impropriety can ruin an analyst’s career
  - Analyst plans security in systems to protect confidential information

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Systems Development Lifecycle

- **System development lifecycle (SDLC，系统开发生命周期)** – the entire process consisting of all activities required to build, launch, and maintain an information system
  - 确定问题和需求： Identify the problem or need and obtain approval
  - 计划和监控项目（做什么、怎么做和谁来做）： Plan and monitor the project
  - 发现和理解问题： Discover and understand the details of the problem or need
  - 设计能解决问题的系统组件： Design the system components that solve the problem or satisfy the need
  - 建立、测试和整合系统组件： Build, test, and integrate system components
  - 完成系统测试与部署： Complete system tests and then deploy the solution

以上为实施系统开发生命周期（SDLC）的 6 个核心过程

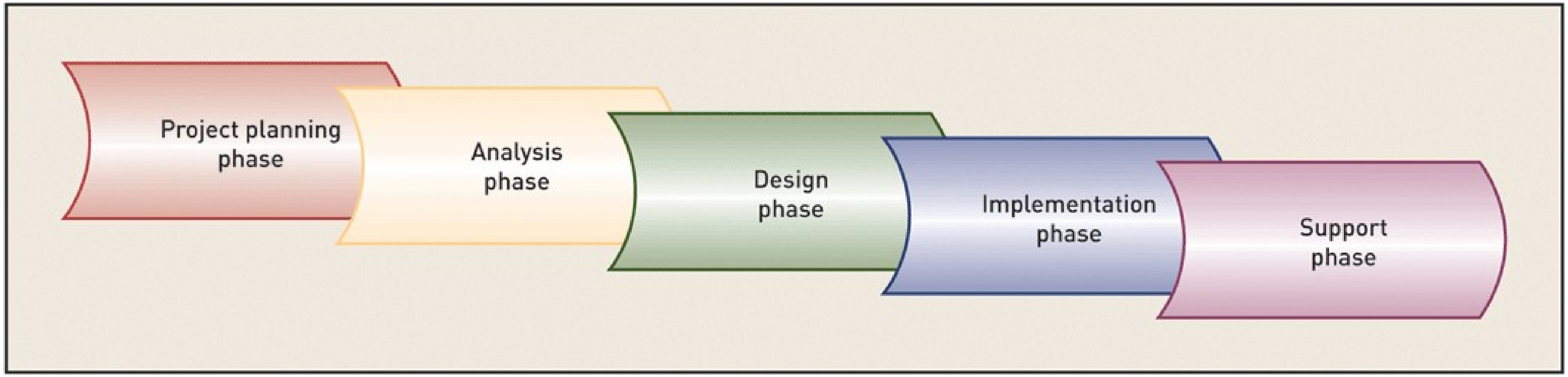
Traditional Predictive Approach to the SDLC

- Project planning （项目计划） — initiate, ensure feasibility, plan schedule, obtain approval for project
- Analysis （分析） — understand business needs and processing requirements
- Design （设计） — define solution system based on requirements and analysis decisions
- Implementation （实施） — construct, test, train users, and install new system
- Support （支持） — keep system running and improve

Information System Development Phases

Figure 2-2

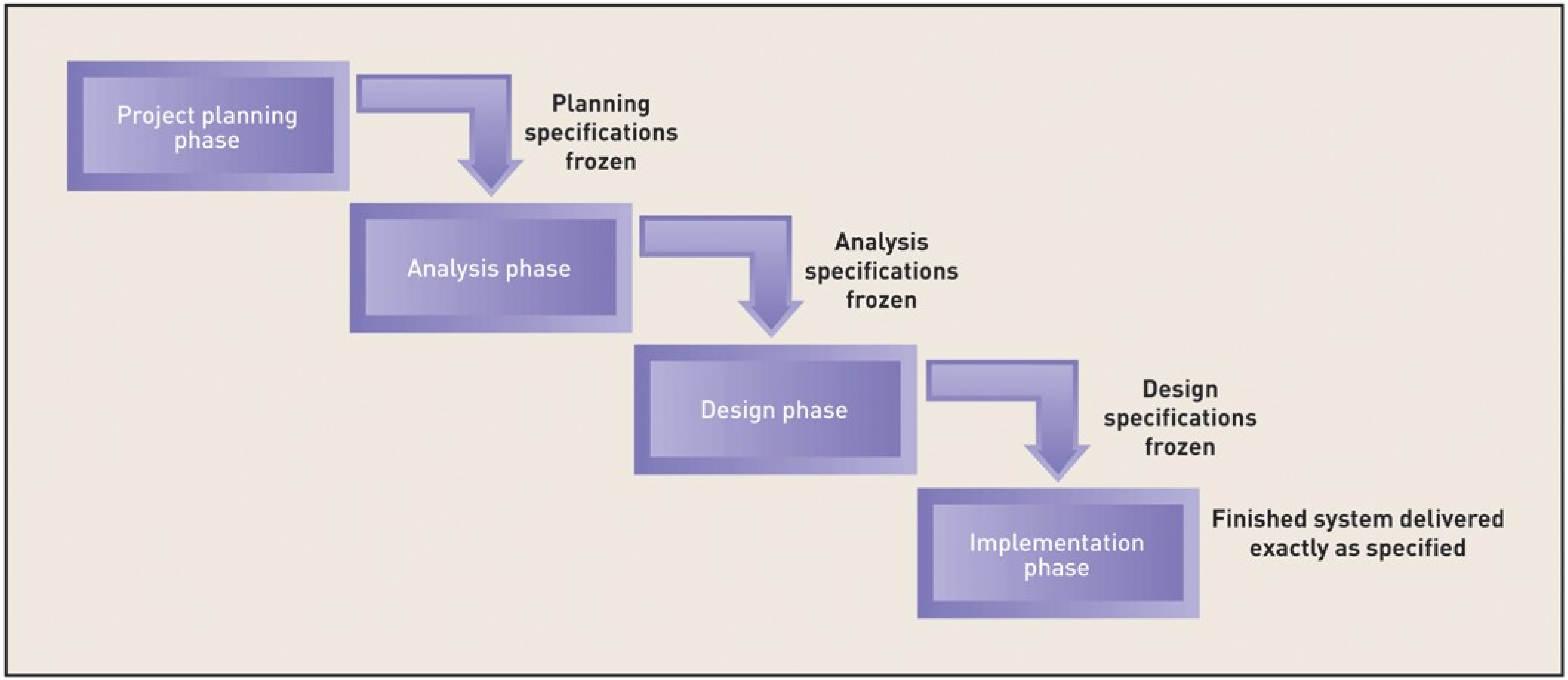
Information system development phases



“Waterfall” Approach to the SDLC： 瀑布式

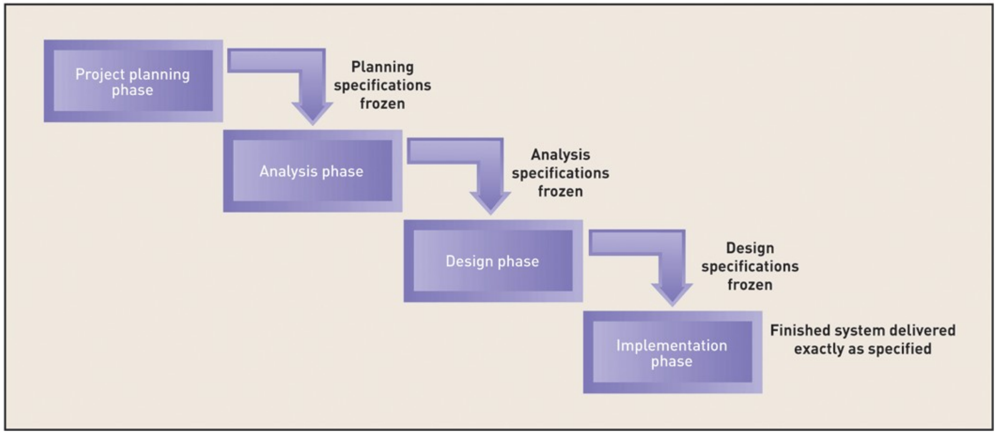
Figure 2-4

The waterfall approach to the SDLC



“Waterfall” Approach to the SDLC： 瀑布式

- “瀑布式”开发方法好不好？可能存在哪些问题？



1）“一蹴而就”：难以适应复杂项目与变化
2）项目进行缓慢，难以快速看到成效

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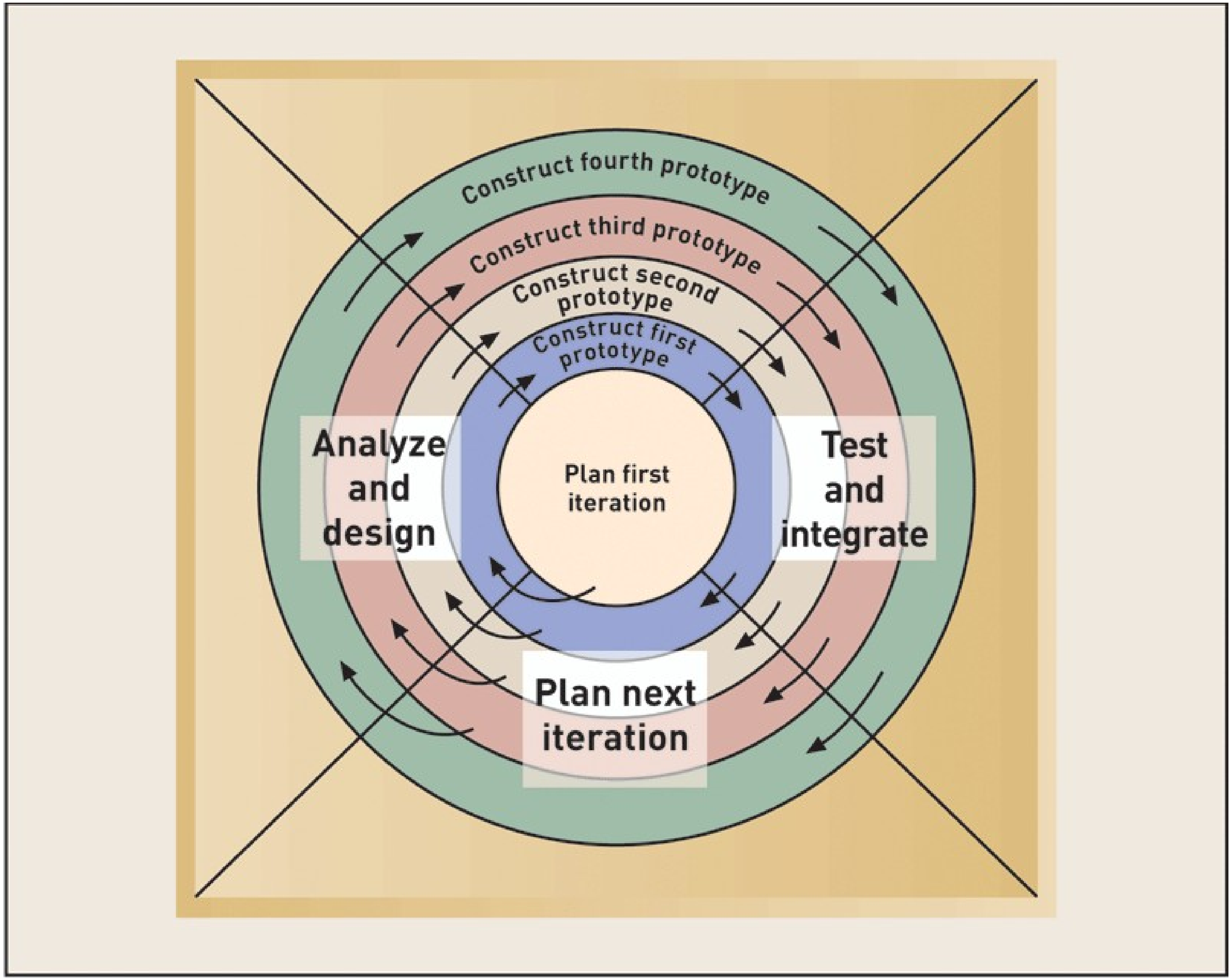
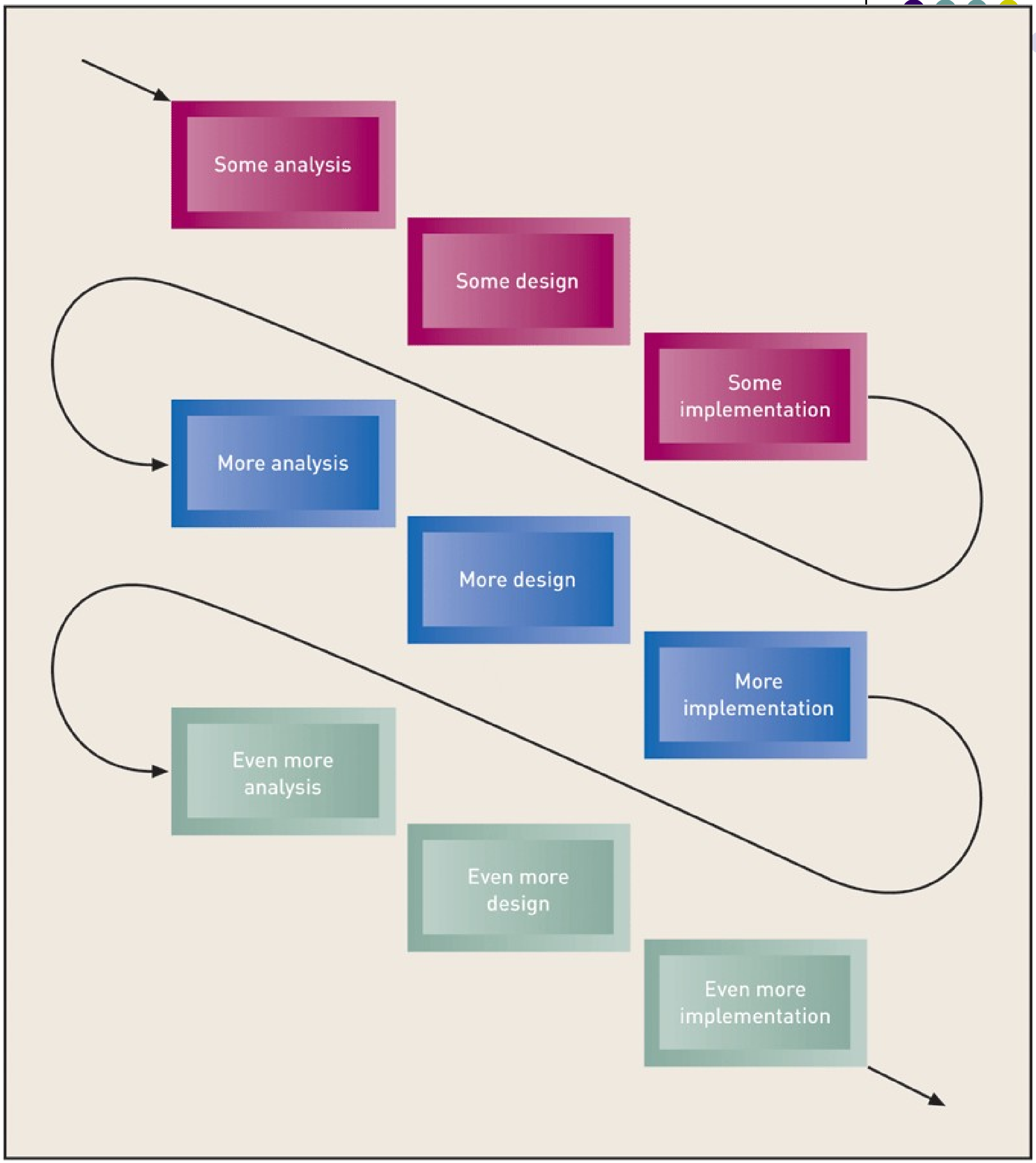
Overview (continued)

- **Agile development** （敏捷开发）— an information system development process that emphasizes flexibility to anticipate new requirements during development
  - Fast on feet; responsive to change
- **Iterative development** （迭代开发） -- an approach to system development in which the system is "grown" piece by piece through multiple iterations
  - Complete small part of system (mini-project), then repeat processes to refine and add more, then repeat to refine and add more, until done

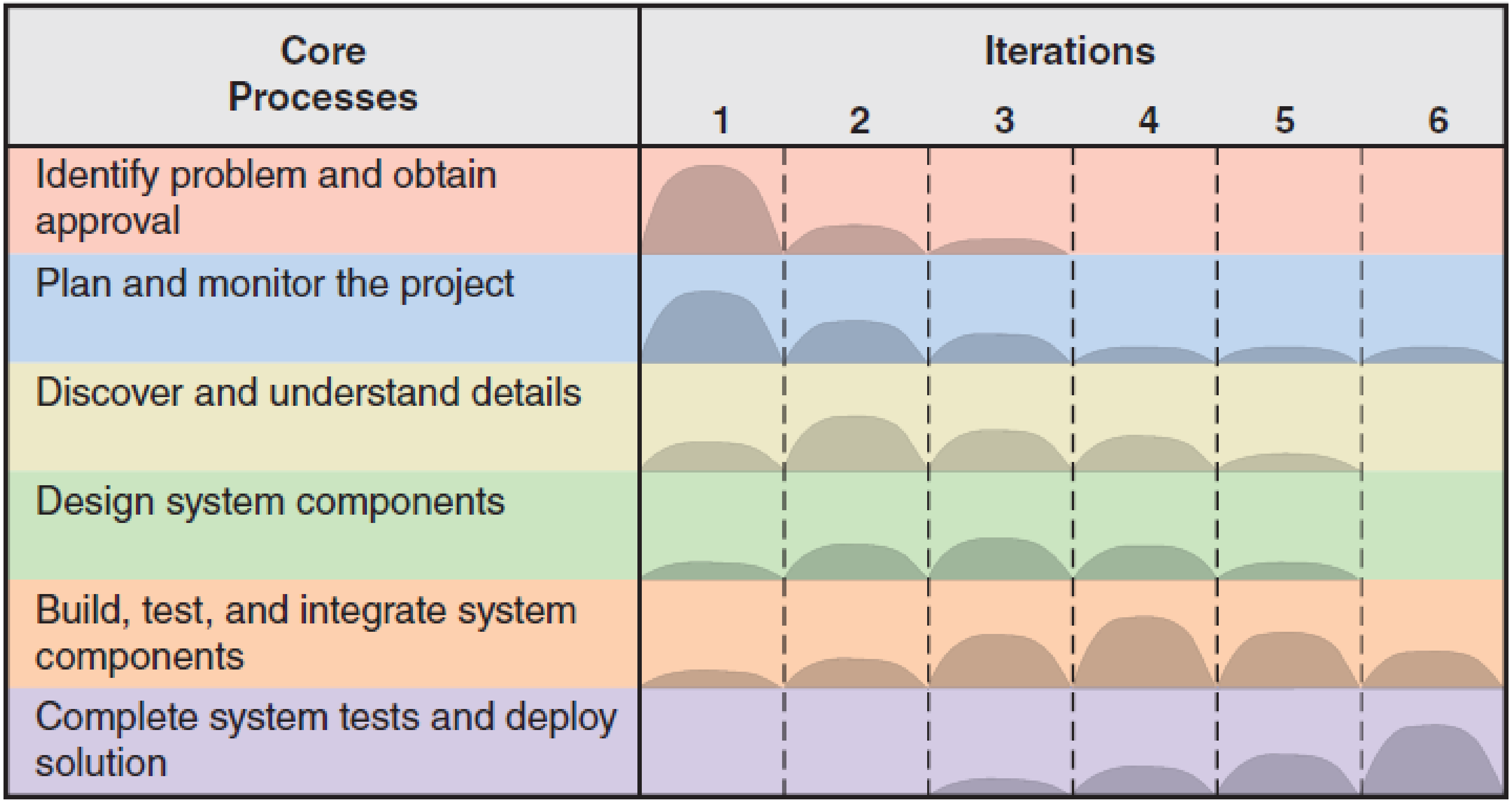
The Spiral Life Cycle Model (螺旋生命周期模型)



Iteration of System Development Activities 系统开发活动的迭代



Iterative and Agile Systems Development Lifecycle (SDLC)



Two Approaches to System Development

- Traditional approach （传统方法）
  - Also called **structured system development （结构化系统开发）**
  - Structured analysis and design technique (SADT，结构化分析与设计）
- Object-oriented approach （面向对象的方法）
  - Views information system as collection of interacting objects that work together to accomplish tasks （将信息系统视为互相作用的对象的集合）

Three Structured Programming Constructs （结构化编程）

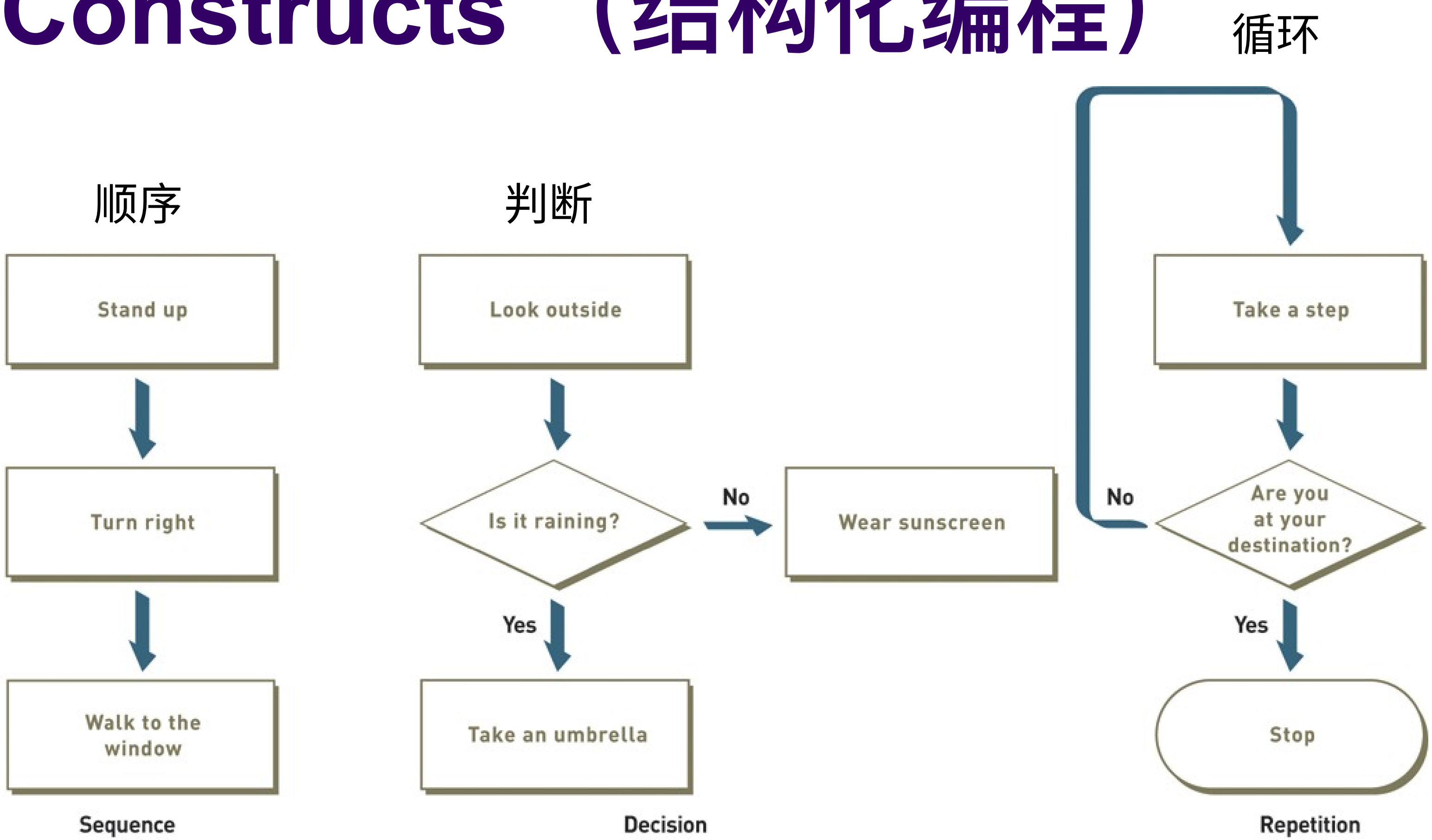


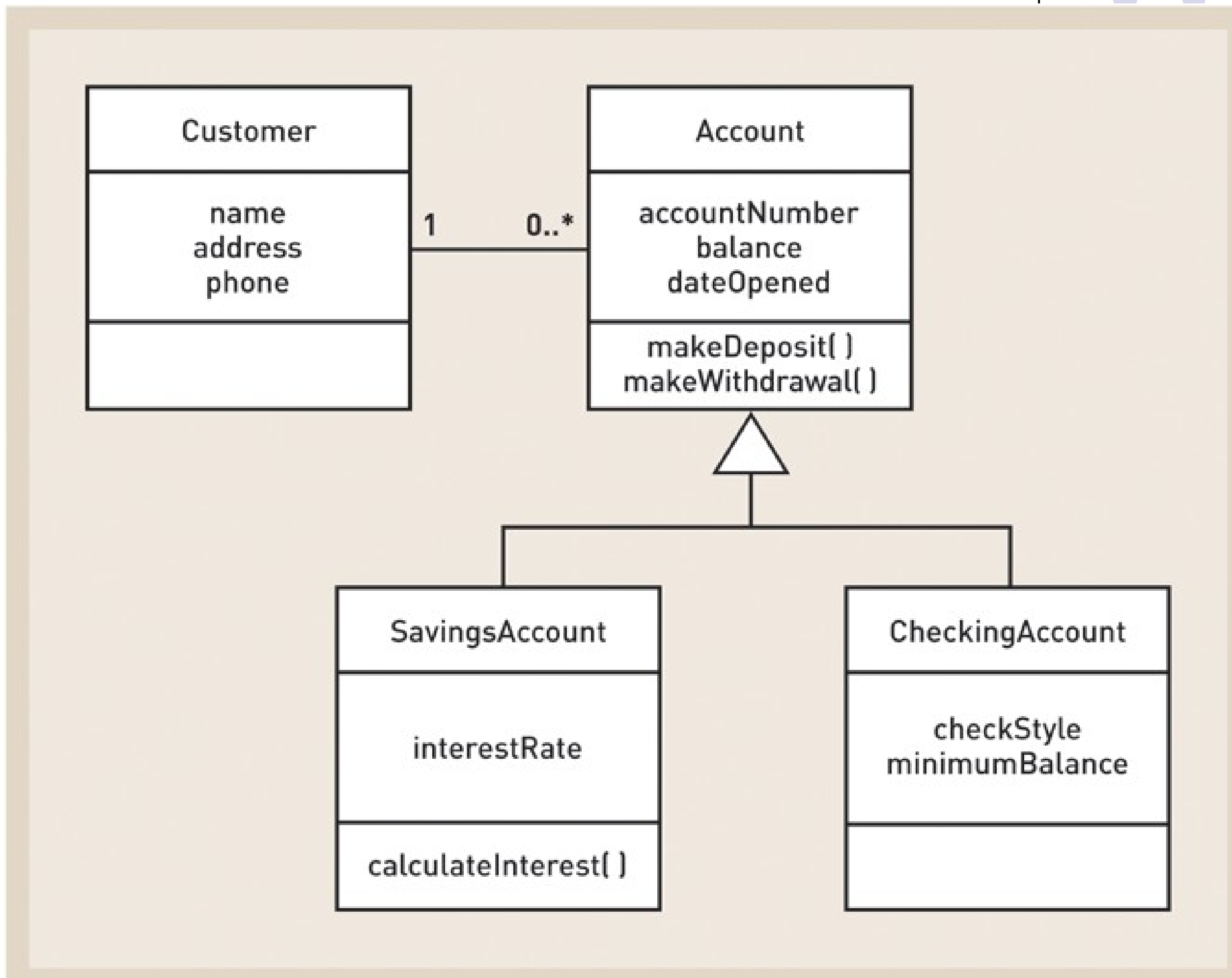
Figure 2-12 Three structured programming constructs

结构化流程图要注意：
1、有始有终
2、单入单出

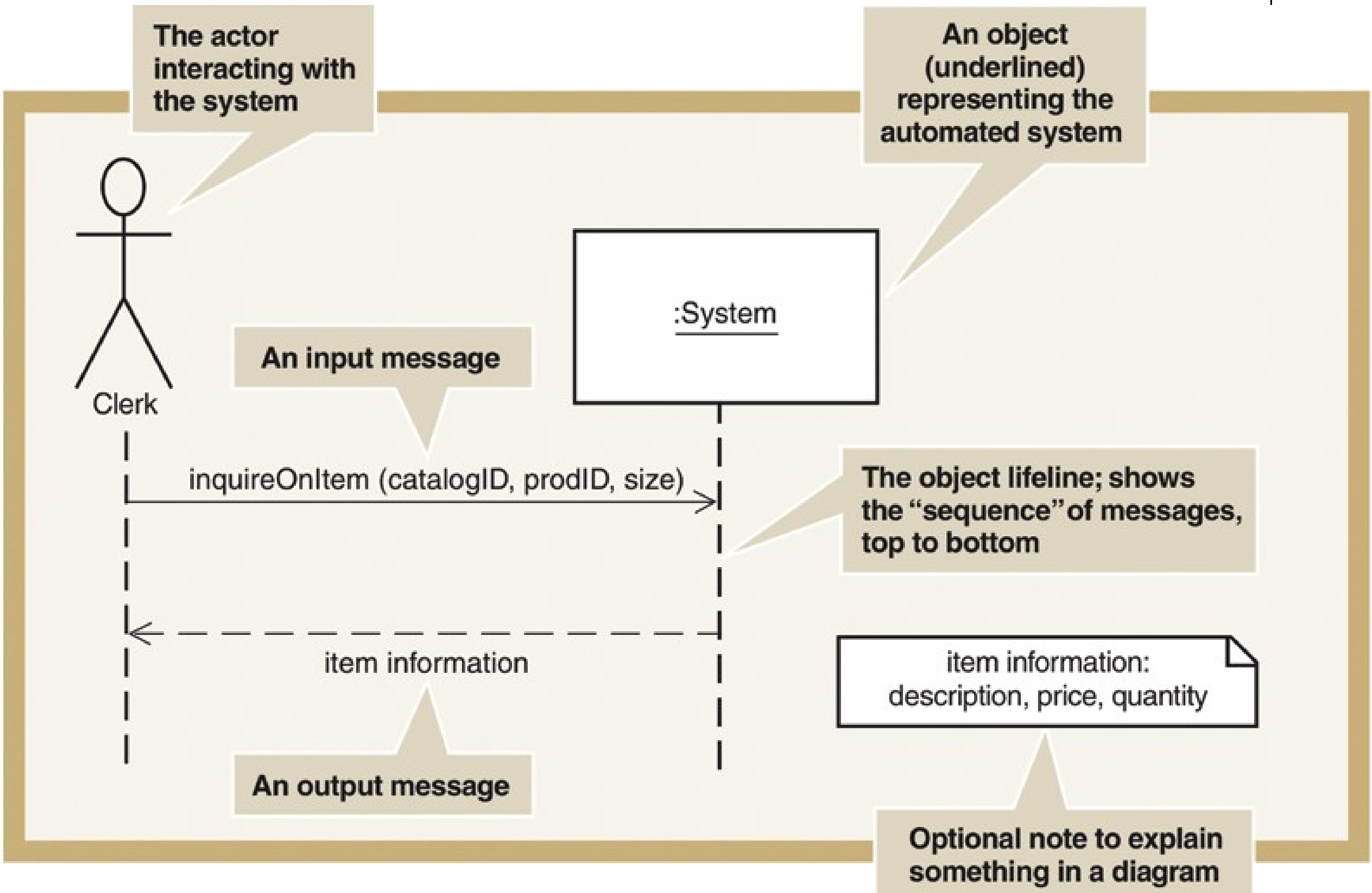
Class Diagram Created During OO Analysis （面向对象）

Figure 2-19

A class diagram created during object-oriented analysis



System Sequence Diagram (SSD， 系统顺序图)



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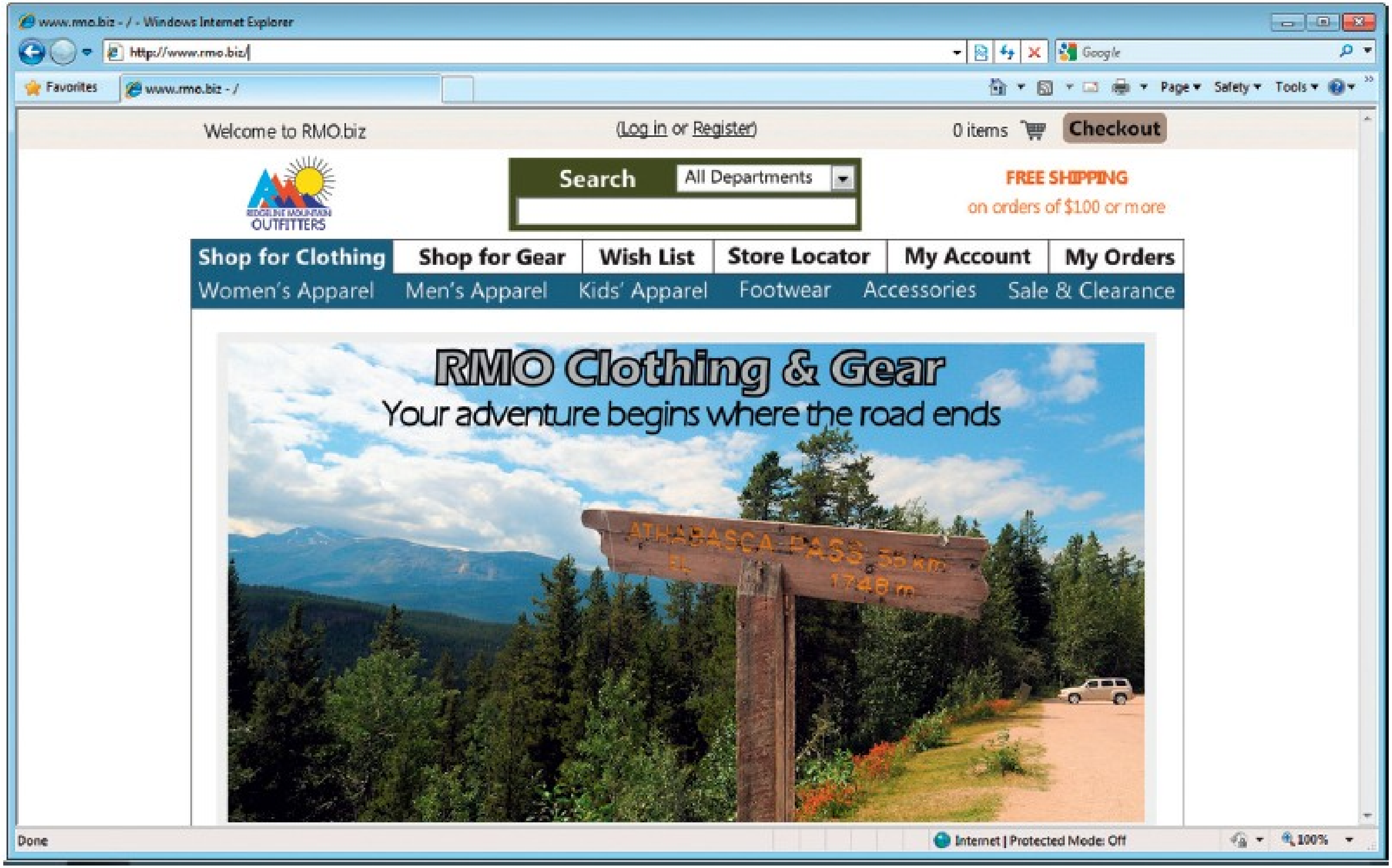
Ridgeline Mountain Outfitters (RMO)

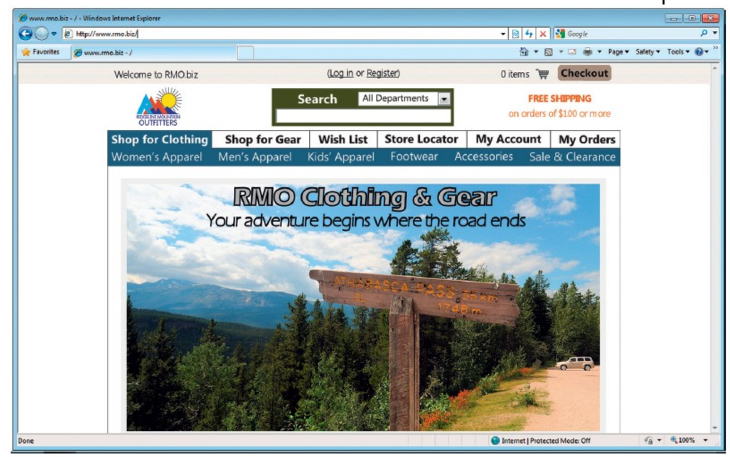
- RMO 是一家大型零售公司，专门生产不同种类的户外运动服装和相关配饰。
  - 最初， RMO 在犹他州等地区把衣服卖给当地的服装店
  - 20 世纪 80 年代后期到 90 年代早期，它通过邮寄和电话订购直接向顾客销售服装
  - 2018 年，它零售商店收入为 6700 万美元，电话订购和邮购达到 1000 万美元，网购销售额达到了 2 亿美元

Ridgeline Mountain Outfitters (RMO)



Ridgeline Mountain Outfitters (RMO)

- 背景:
  - 为了保证生产线的现代化， RMO 的采购代理人会参加世界各地的服装展和面料展。



  - 过去，这些代理人想要订购 RMO 的商品时，需要在展览中与销售人员交换联系信息，等回到办公室后在通过电子邮件和电话进一步制定合同和购买订单。
  - 为了迅速完成订单， RMO 启动了一个项目：开发一个系统来收集和跟进它的代理人信息和待销售商品

RMO Tradeshow System

- **Problem**-- purchasing agents attend apparel and fabric trade shows around the world to order new products from suppliers （采购代理从供应商处订购新产品）
- **Need**– information system (app) to collect and track information about suppliers and new products while at tradeshows （收集供应商和产品信息）
- **Tradeshow Project**– is proposed
  - Supplier information subsystem （供应商信息子系统）
  - Product information subsystem （产品信息子系统）

Pre-Project Activities

- Identify the problem and document the objective of the system (core process 1)
  - Preliminary investigation （初步调查）
  - System Vision Document （系统愿景文档）
- Obtain approval to commence the project (core process 1)
  - Meet with key stakeholders, including executive management （与利益相关者会面）
  - Decision reached, approve plan and budget （做出计划与预算决定）
    - Core process 1：确定问题和需求： Identify the problem or need and obtain approval

System Vision Document

Problem description：为什么要做?

System capabilities：做什么?

Business benefits：能带来什么收益?

System Vision Document
RMO Tradeshow System

Problem Description

Trade shows have become an important information source for new products, new fashions, and new fabrics. In addition to the large providers of outdoor clothing and fabrics, there are many smaller providers. It is important for RMO to capture information about these suppliers while the trade show is in progress. It is also important to obtain information about specific merchandise products that RMO plans to purchase. Additionally, if quality photographs of the products can be obtained while at the trade show, then the creation of online product pages is greatly facilitated.

It is recommended that a new system be developed and deployed so field purchasing agents can communicate more rapidly with the home office about suppliers and specific products of interest. This system should be deployed on portable equipment.

System Capabilities

The new system should be capable of:

- Collecting and storing information about the manufacturer/wholesaler (suppliers)
- Collecting and storing information about sales representatives and other key personnel for each supplier
- Collecting information about products
- Taking pictures of products (and/or uploading stock images of products)
- Functioning as a stand-alone without connection
- Connecting via Wi-Fi (Internet) and transmitting data
- Connecting via telephone and transmitting data

Business Benefits

It is anticipated that the deployment of this new system will provide the following business benefits to RMO:

- Increase timely communication between trade show attendees and home office, thereby improving the quality and speed of purchase order decisions
- Maintain correct and current information about suppliers and their key personnel, thereby facilitating rapid communication with suppliers
- Maintain correct and rapid information and images about new products, thereby facilitating the development of catalogs and Web pages
- Expedite the placing of purchase orders for new merchandise, thereby catching trends more rapidly and speeding up product availability

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为什么要做?
区域代理商能够更快速地与供应商和感兴趣的产品进行沟通

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- Functioning as a stand-alone without connection
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做什么?
收集和记录信息（供应商关键信息，产品信息），通过平台来沟通

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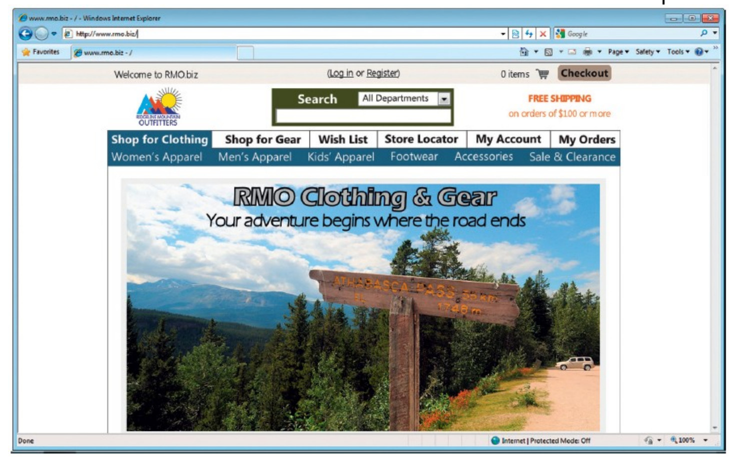
能带来什么收益?
维护客户关系，更好地管理订单

Ridgeline Mountain Outfitters (RMO)

- “RMO 贸易展览系统”立项了！



- 开启第 1 个迭代：
  - 我们计划用 6 天时间完成本次迭代，来展示 SDLC 的所有核心过程

Day 1 Activities

- Core Process 2: Plan the Project （计划项目）
  - Determine the major components (functional areas) that are needed （确定主要组件）
    - Supplier information subsystem （供应商信息子系统）
    - Product information subsystem （产品信息子系统）
  - Define the iterations and assign each function to an iteration （将每个功能分配给每个迭代）
    - Decide to do Supplier subsystem first （先做供应商子系统）
    - Plan one iteration as it is small and straight forward
  - Determine team members and responsibilities （确定团队成员和职责）

Work Breakdown Structure for Iteration （工作分解结构，基于 SDLC 后四个核心过程）

1. 发现并理解问题各个方面的细节
2. 设计问题解决方案的组成部分（界面，数据库，结构等）
3. 代码实现各个组件
4. 进行系统级测试和部署

Work Breakdown Structure

I. Discover and understand the details of all aspects of the problem.

1. Meet with the Purchasing Department manager. ~ 3 hours
2. Meet with several purchasing agents. ~ 4 hours
3. Identify and define use cases. ~ 3 hours
4. Identify and define information requirements. ~ 2 hours
5. Develop workflows and descriptions for the use cases. ~ 6 hours

II. Design the components of the solution to the problem.

1. Design (lay out) input screens, output screens, and reports. ~ 8 hours
2. Design and build database (attributes, keys, indexes). ~ 4 hours
3. Design overall architecture. ~ 4 hours
4. Design program details. ~ 6 hours

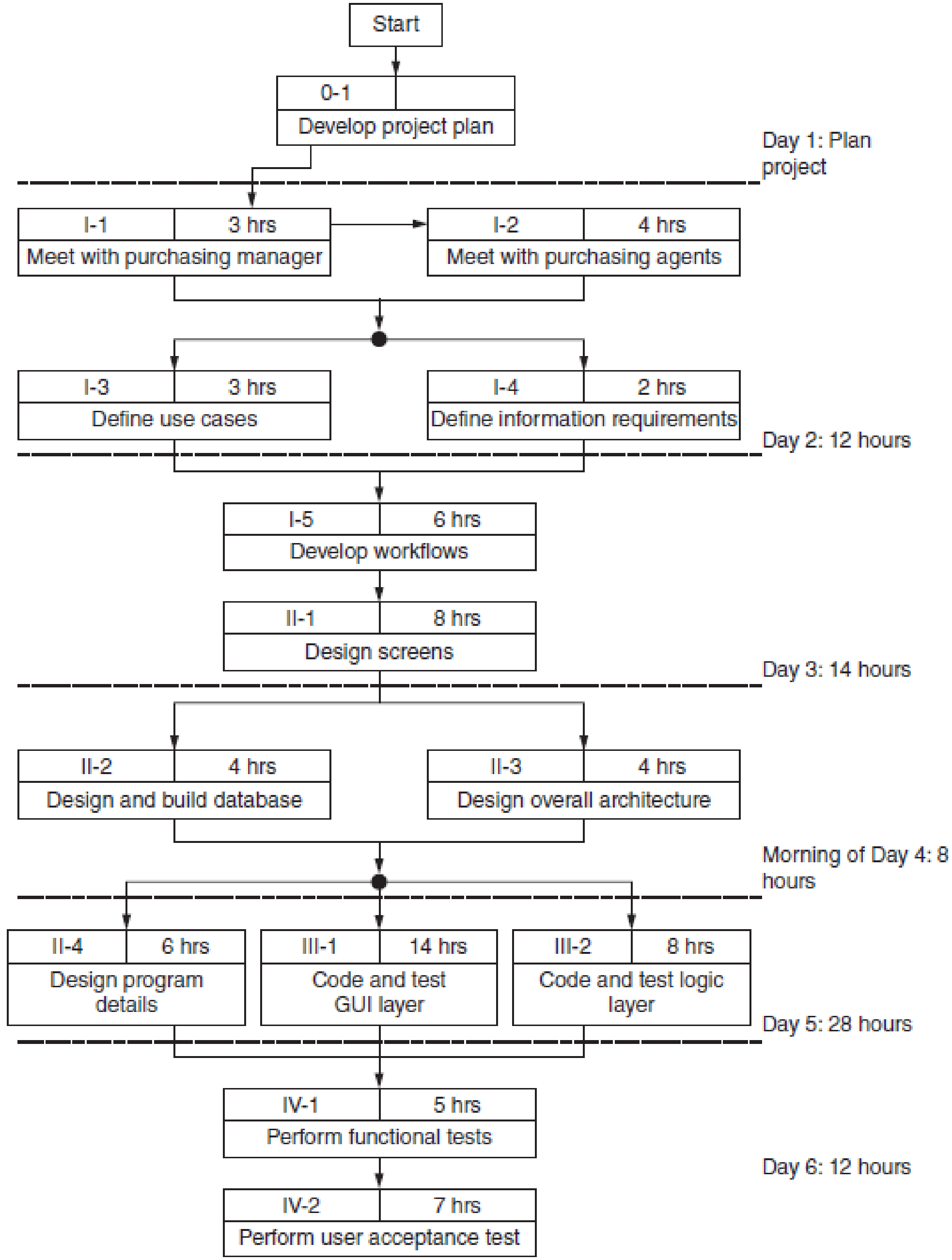
III. Build the components and integrate everything into the solution.

1. Code and unit test GUI layer programs. ~ 14 hours
2. Code and unit test Logic layer programs. ~ 8 hours

IV. Perform all system-level tests and then deploy the solution.

1. Perform system functionality tests. ~ 5 hours
2. Perform user acceptance test. ~ 8 hours

Work Sequence Draft for Iteration（工作顺序草案）



Elaborates on Work Breakdown Structure（详细阐述工作分解结构）

Work Sequence Draft for Iteration （工作顺序草案）

- 三重优点：
  - 它帮助团队组织工作，在编程开始之前就有足够的时间来考虑如何设计。
  - 它提供了一个衡量标准来看这个迭代是否符合日程要求。
  - 如果项目可能会延误，那么项目的领导者能够提早开始组织资源来帮助完成项目。

Day 2 Activities

- Core Process 3: Discover and Understand Details （发现和理解细节）
  - Do preliminary fact-finding to understand requirements （了解需求）
  - Develop a preliminary list of use cases （用例） and a use case diagram （用例图）
  - Develop a preliminary list of classes （类） and a class diagram （类图）

Day 2 Activities

- Core Process 3: Discover and Understand Details （发现和理解细节）
  - Do preliminary fact-finding to understand requirements （了解需求）
    - 采访核心用户
    - 观察工作过程
    - 回顾现存的文件和系统
    - 调查其他公司和其他系统

Day 2 Activities

- Core Process 3: Discover and Understand Details
  - Do preliminary fact-finding to understand requirements
  - Develop a preliminary list of use cases （用例） and a use case diagram （用例图）
    - 用例记录了一个简单的用户触发商业活动和系统对这个活动的回应

Identify Use Cases

Both subsystems

动词短语

Use Case	Description
Look up supplier	Using supplier name, find supplier information and contacts
Enter/update supplier information	Enter (new) or update (existing) supplier information
Look up contact	Using contact name, find contact information
Enter/update contact information	Enter (new) or update (existing) contact information
Look up product information	Using description or supplier name, look up product information
Enter/update product information	Enter (new) or update (existing) product information
Upload product image	Upload images of the merchandise product

Day 2 Activities

- Core Process 3: Discover and Understand Details
  - Do preliminary fact-finding to understand requirements
  - Develop a preliminary list of use cases and a use case diagram
  - Develop a preliminary list of classes （类） and a class diagram （类图）
    - 寻找能描述事物类别的名词

Identify Object Classes

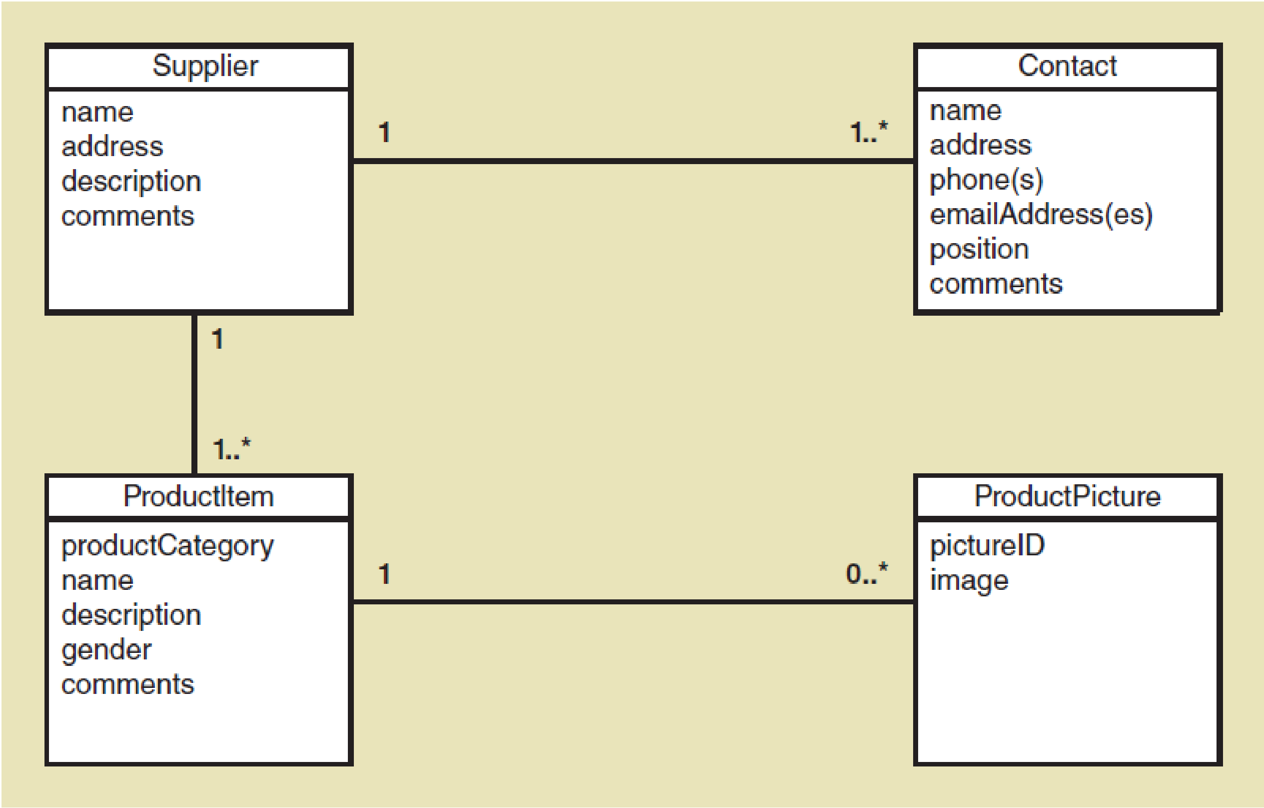
Both subsystems

Object Classes	Attributes
Supplier 名词	supplier name, address, description, comments
Contact	name, address, phone(s), e-mail address(es), position, comments
Product	category, name, description, gender, comments
ProductPicture	ID, image

Preliminary Class Diagram

Both subsystems

除了提供对象类列表外，分析员经常会绘制一个类、属性和它们与另外的类之间的关系图，称为类图：



Day 3 Activities

- Core Process 3: Discover and Understand Details （发现和理解细节）
  - Do in-depth fact-finding to understand requirements
  - Understand and document the detailed workflow of each use case
- Core Process 4: Design System Components （设计系统组件）
  - Define the user experience with screens and reports （用屏幕展示和报告给用户直观体验）

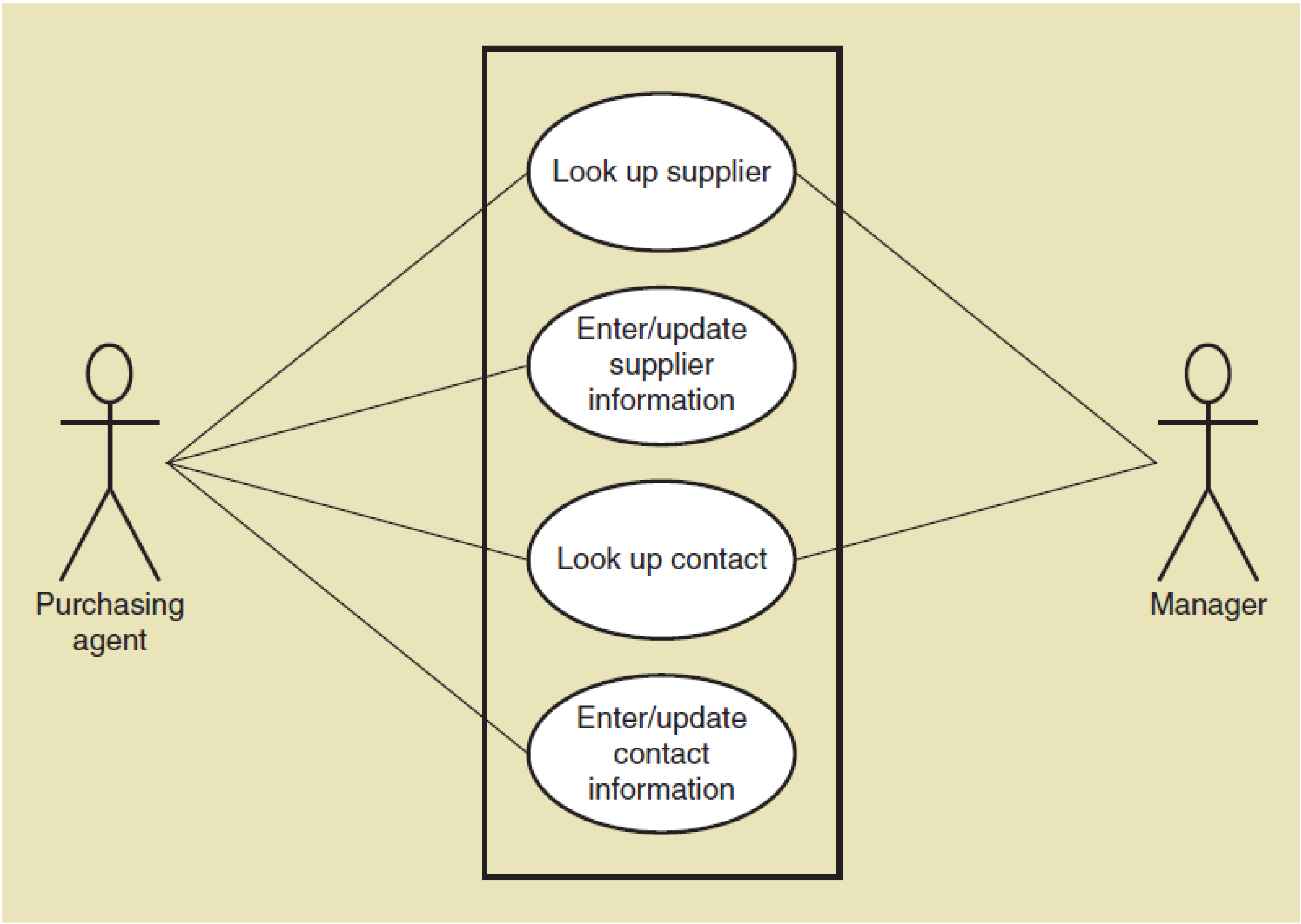
Day 3 Activities

- Core Process 3: Discover and Understand Details （发现和理解细节）
  - Do in-depth fact-finding to understand requirements （事实调查）
  - Understand and document the detailed workflow of each use case （了解用例工作流程）
- Core Process 4: Design System Components
  - Define the user experience with screens and reports

Details Focus on Supplier Information Subsystem （供应商信息子系统细节）

- Use cases:
  - Look up supplier （查找供应商）
  - Enter/update supplier information （输入 / 更新供应商信息）
  - Lookup contact information （查找联系信息）
  - Enter/update contract information （输入 / 更新合同信息）

Use Case Diagram （用例图）
Supplier information subsystem

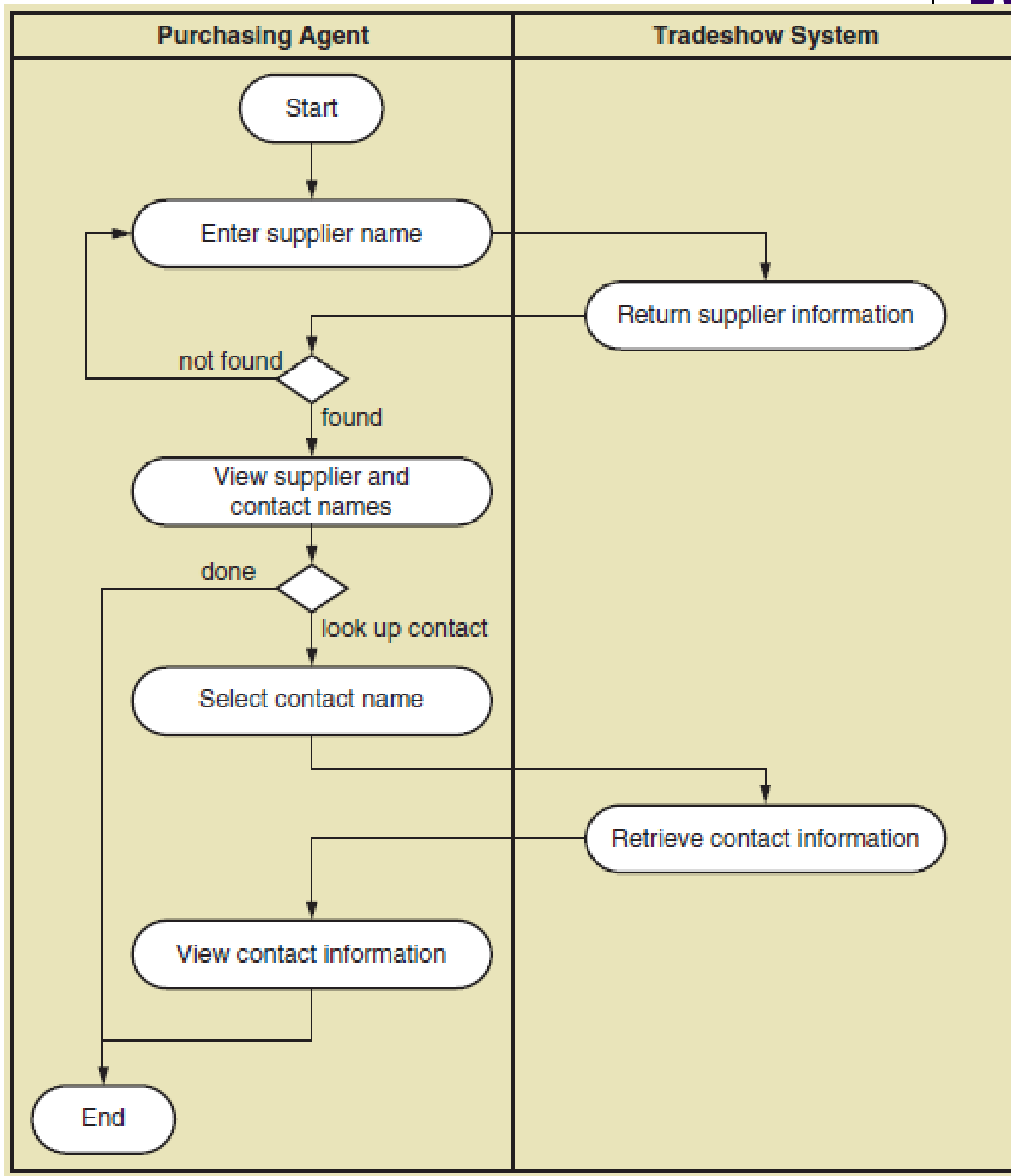


Activity Diagram (Workflow)

开发工作流图：
记录一个用例的细节

Look up supplier use case

1. 椭圆代表任务，菱形代表决策点
2. 穿越中心线的箭头代表了系统和用户之间的相互作用

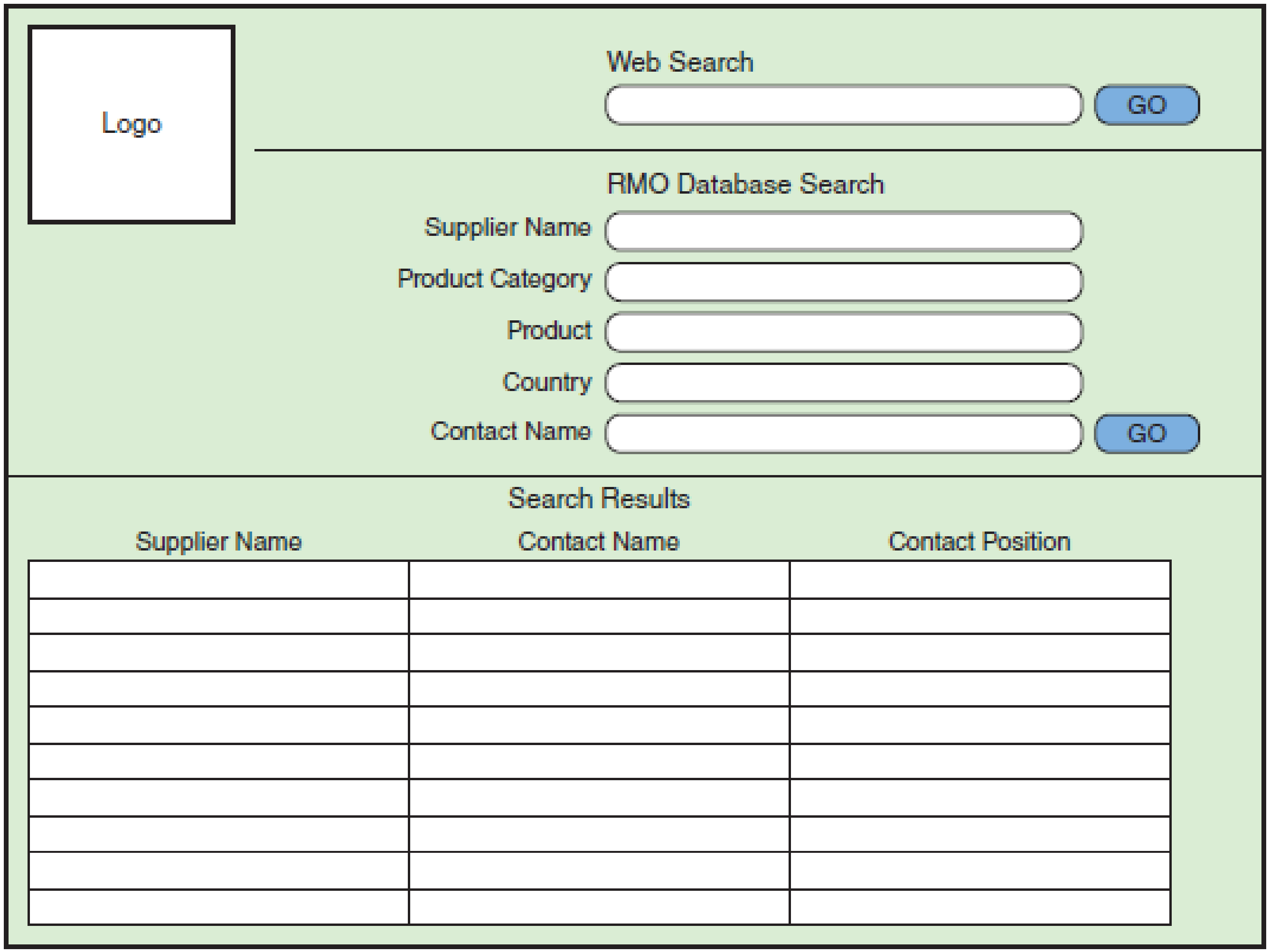


Day 3 Activities

- Core Process 3: Discover and Understand Details
  - Do in-depth fact-finding to understand requirements
  - Understand and document the detailed workflow of each use case
- Core Process 4: Design System Components （设计系统组件）
  - Define the user experience with screens and reports （用屏幕和报告定义用户体验）

Draft Screen Layout

Look up supplier use case



Day 4 Activities

- Core Process 4: Design System Components
  - Design the database (设计数据库)
    - Table design （表）
    - Key and index identification （主键、索引）
    - Attribute types （属性）
    - Referential integrity （参照完整性）
  - Design the system’s high level structure （高层结构）
    - Browser, Windows, or Smart phone （网页 / 单机 / 移动环境）
    - Architectural configuration (组件)
    - Design class diagram （设计类图）
    - Subsystem architectural design （系统结构）

Database Schema

Table Name	Attributes
Supplier	SupplierID: integer {key} Name: string {index} Address1: string Address1: string City: string State-province: string Postal-code: string Country: string SupplierWebURL: string Comments: string
Contact	ContactID: integer {key} SupplierID: integer {foreign key} Name: string {index} Title: string WorkAddress1: string WorkAddress2: string WorkCity: string WorkState: string WorkPostal-code: string WorkCountry: string WorkPhone: string MobilePhone: string EmailAddress1: string EmailAddress2: string Comments: string

Day 4 Activities

- Core Process 4: Design System Components
  - Design the database (设计数据库)
    - Table design （表）
    - Key and index identification （主键、索引）
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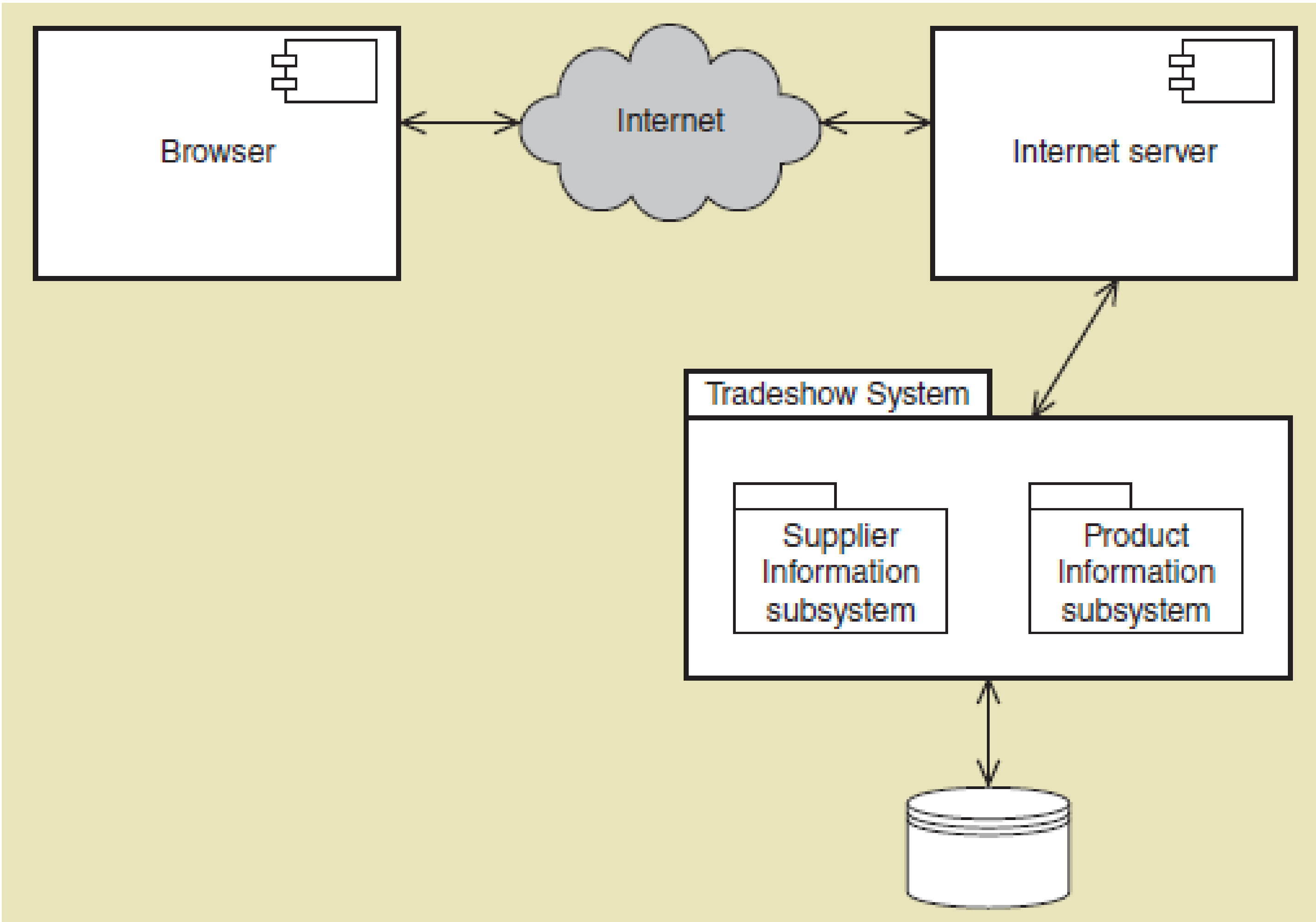
    - Subsystem architectural design （系统结构）

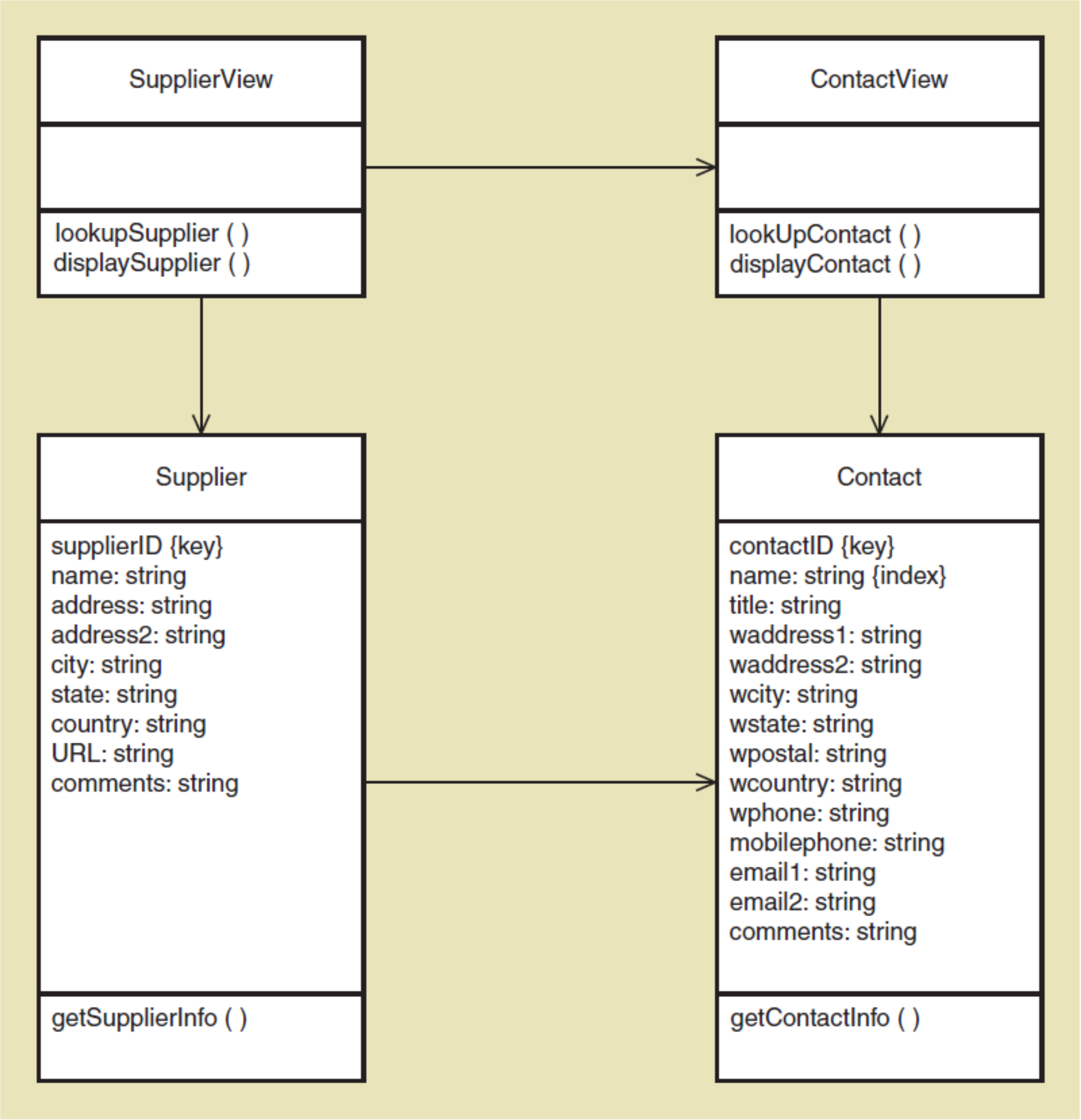
Architectural Configuration Diagram



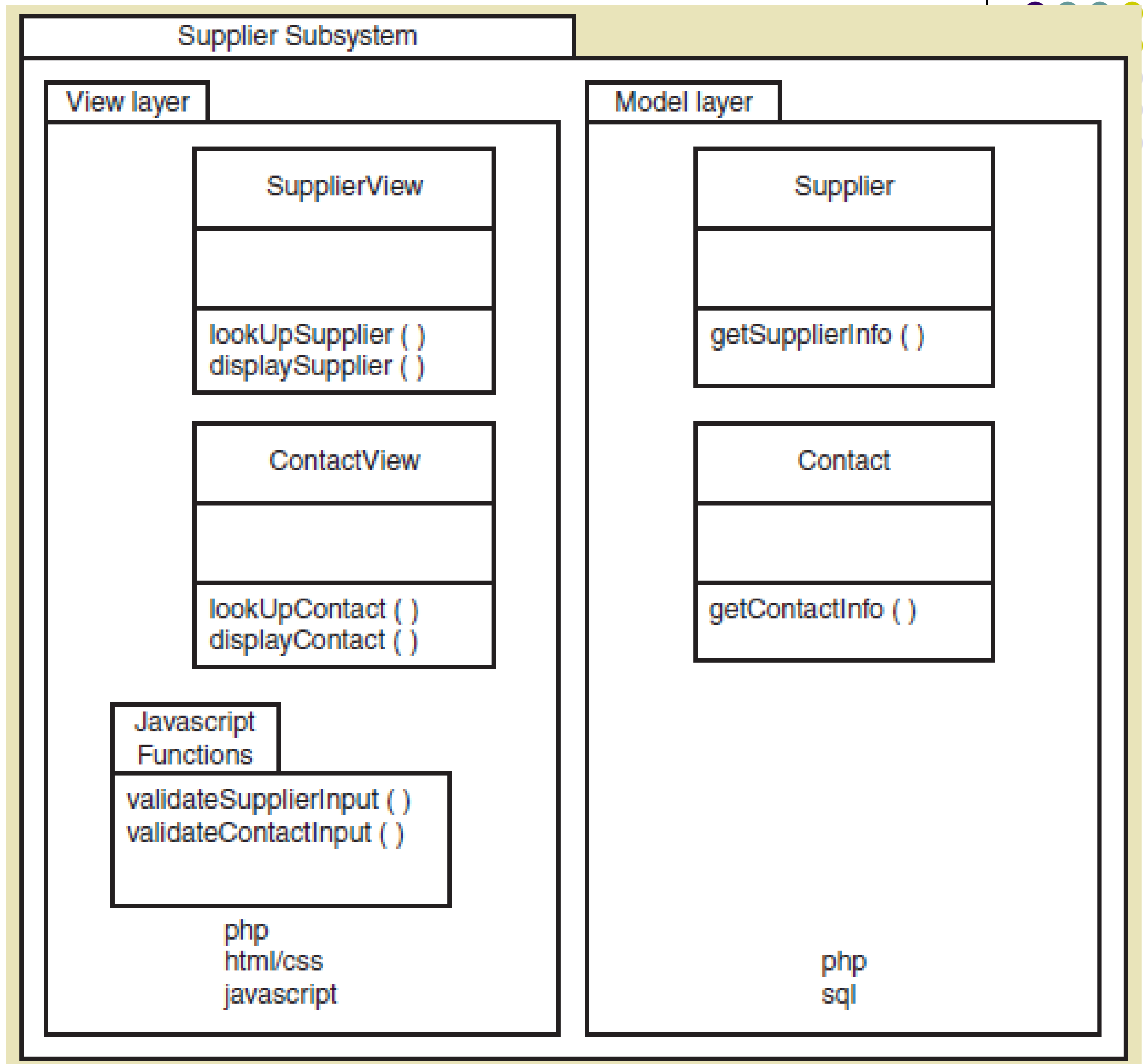
Preliminary Design Class Diagram

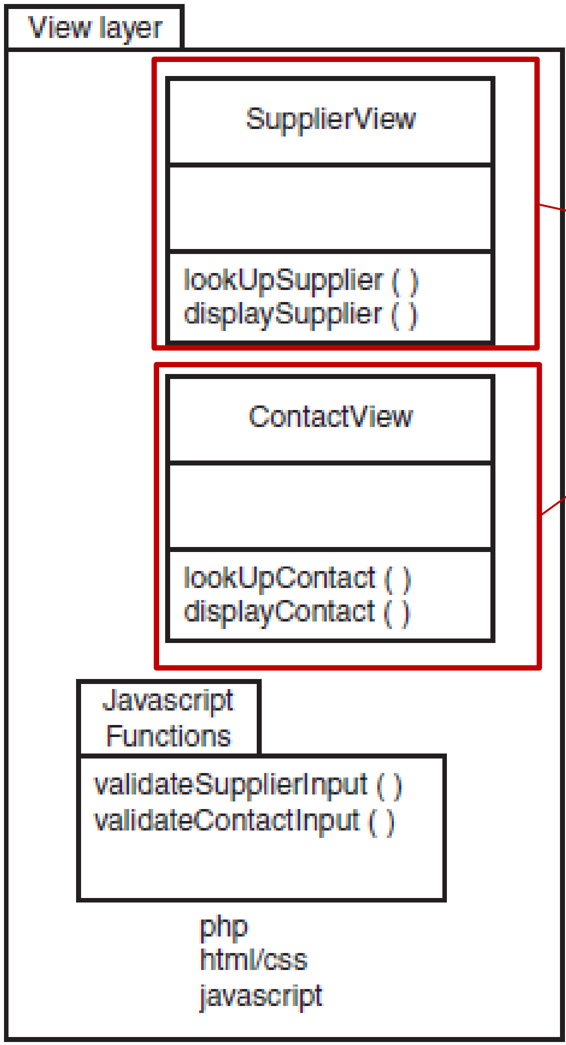
Includes View Layer （视图层） Classes and Domain Layer （域层） Classes

Need to add Utility Classes （工具类） as well



Subsystem Architectural Design Diagram （子系统架构设计图）






```
<?php
    class SupplierView
    {
        private Supplier $theSupplier;

        function __construct()
        {
            $this->theSupplier = new Supplier();
        }

        function lookupSupplier()
        {
          include('lookupSupplier.inc.html');
        }

        function displaySupplier()
        {
              include('displaySupplierTop.inc.html');
              extract($_REQUEST);    // get Form data
            //Call Supplier class to retrieve the data
            $results = $theSupplier->getSupplierInfo($supplier, $category,
                                                $product, $country, $contact);

           foreach ($results as $resultItem){
           ?>
                   <tr>
                       <td style="border:1px solid black">
                              <?php echo $resultItem->supplierName?></td>
                       <td style="border:1px solid black">
                               <?php echo $resultItem->contactName?></td>
                       <td style="border:1px solid black">
                           <?php echo $resultItem->contactPosition?></td>
                   </tr>
            <?php }
            include('displaySupplierFoot.inc.html');
        }
    }
?>
```

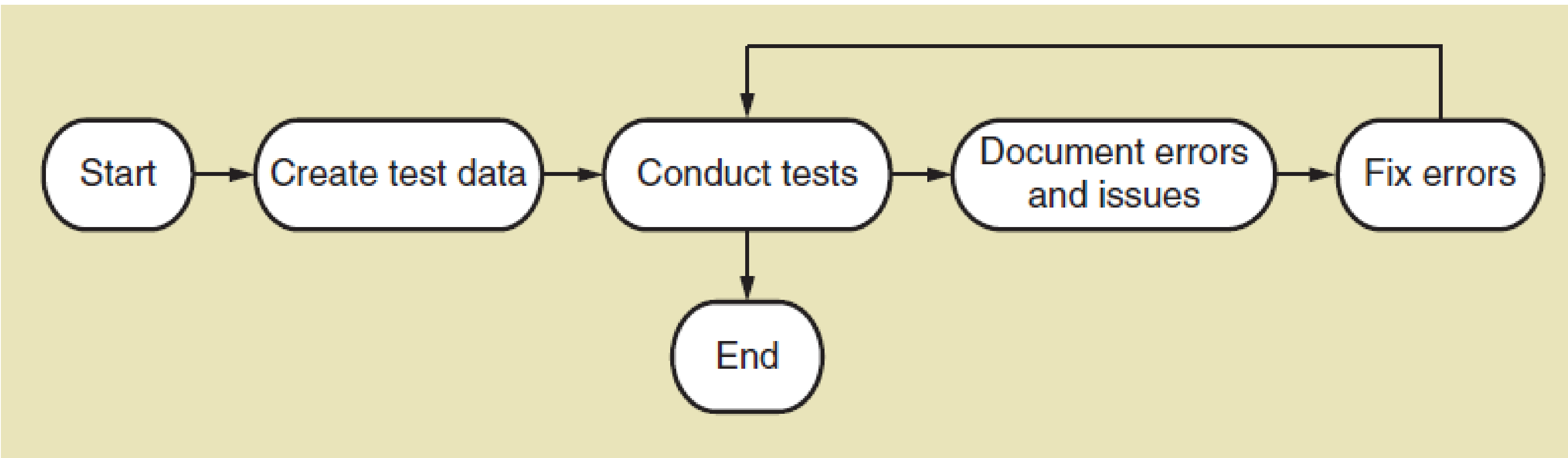
Notes on Managing the Project (项目管理须知)

- Lots of design diagrams shown （设计图表展示了）
  - Design in a complex activity with multiple levels
    - High level architectural （高级架构）
    - Low level detailed design （底层详细设计）
  - One diagram builds on/complements another （一个图表建立在另一个图表的基础上）
- Programming is also done concurrently （编程是并发完成的）
  - You don’t design everything then code （不需要设计所有东西再编码）
  - You do some design, some coding, some design, some coding （设计，编码，设计，编码迭代循环）

Day 5 Activities

- Core Process 4: Design System Components （设计系统组件）
  - Continue with design details （继续设计细节）
  - Proceed use case by use case （推进用例）
- Core Process 5: Build, Test, and Integrate System Components （构建、测试和集成系统组件）
  - Continue programming (继续编程)
  - Build use case by use case （逐个用例来构建）
  - Perform unit and integration tests （执行单元和集成测试）

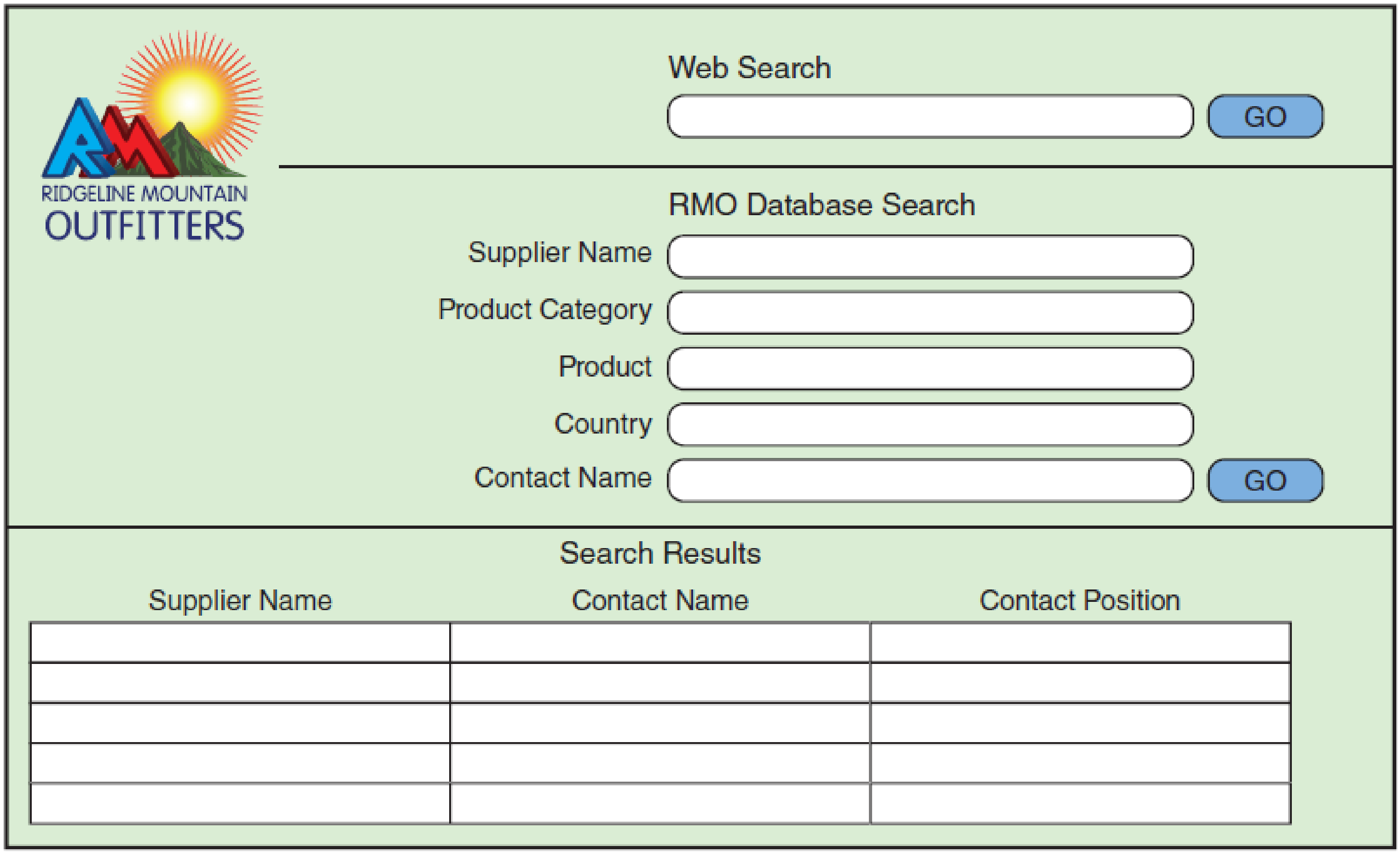
Workflow of Testing Tasks
测试任务流程



Day 6 Activities

- Core Process 6: Complete System Testing and Deploy System （完成系统测试和部署系统）
  - Perform system functional testing （执行系统功能测试）
  - Perform user acceptance testing （执行用户验收测试）
  - Possibly deploy part of system （可能部署系统的一部分进行测试）

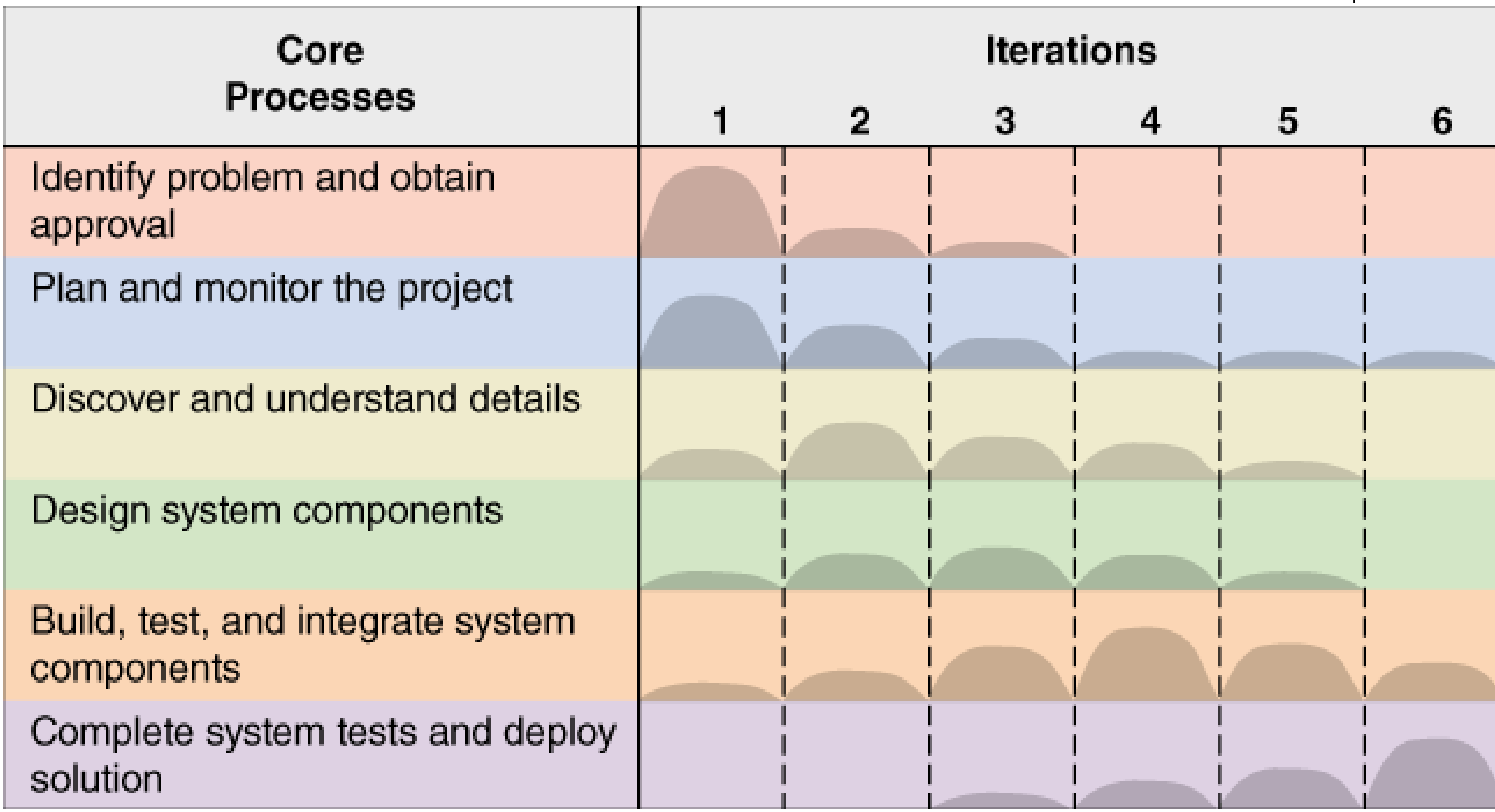
Screen Capture for Look up supplier use case （查找供应商界面）



First Iteration Recap （第一次迭代回顾）

- This was a 6 day iteration of small project
  - Most iterations are longer (2 to 4 weeks)
  - This project might be 2 iterations
  - Most projects have many more iterations （大多数项目都有更多迭代）
- End users need to be involved, particularly in day 1, 2, 3 and 6. （终端用户需要参与进来）
- Days 4 and 5 involved design and programming <u>concurrently</u>. （设计与编程同步进行）
  - Lots of time was spent programming along with design (not emphasized here)

This Book is about Activities and Tasks in the SDLC



Chapter 1 Outline

- Software Development and Systems Analysis and Design
- Systems Development Lifecycle
- Iterative Development
- Introduction to Ridgeline Mountain Outfitters
- Developing RMO’s Tradeshow Systems
- Where You are Headed—The Rest of the Book （书的其余章节）

Where You Are Headed

- Chapter 1: From Beginning to End （从开始到结束）
  - Small project overview emphasizing analysis and design and iterative development （系统开发生命周期）
  - Done!
- Chapter 2: Investigating System Requirements （需求）
  - More about core process 3: Systems analysis activities
- Chapter 3: Use Cases （用例）
  - Techniques for Identifying and modeling use cases for systems analysis

Where You Are Headed

- Chapter 4: Domain Modeling （域建模）
  - Techniques for Identifying and modeling domain classes for systems analysis
- Chapter 5: Extending the Requirements Models （扩展需求模型）
  - Modeling more details about use cases and domain classes for systems analysis

Where You Are Headed

- Chapter 6: Essentials of Design （设计）
  - More about core process 4: system design activities
- Chapter 7: Designing User and System Interfaces （接口）
  - Human computer interaction, user interface design principles, outputs and reports, system interfaces
- Chapter 8: Approaches to System Development （系统开发）
  - More about the SDLC, models, tools, techniques, and agile methodologies
- Chapter 9: Project Planning and Project Management （项目计划和项目管理）

Where You Are Headed

- Chapter 10: Object-Oriented Design: Principles （面向对象设计原则）
  - Design principles, design models, and designing use cases
- Chapter 11: Object-Oriented Design: Use Case Realization （面向对象设计）
  - Three layer design and design patterns
- Chapter 12: Databases, Controls, and Security （数据库、控件和安全性）
  - More about database design and protecting the integrity of the system.

Summary

- This text is about developing information systems that solve an organization need
- Chapter 1 takes you through the whole process for one small information system
- System development involves 6 core processes, known as the SDLC （6 个核心过程）
- The rest of the text elaborates on the basic processes shown in chapter 1

Summary

- Terms to review and know include:
  - Computer application （计算机应用）
  - Information system （信息系统）
  - Systems analysis （系统分析）
  - System design （系统设计）
  - System development lifecycle (SDLC， 系统开发生命周期）
  - Information system development process (methodology， 信息系统开发过程）
  - Agile development （敏捷开发）
  - Iterative development （迭代开发）